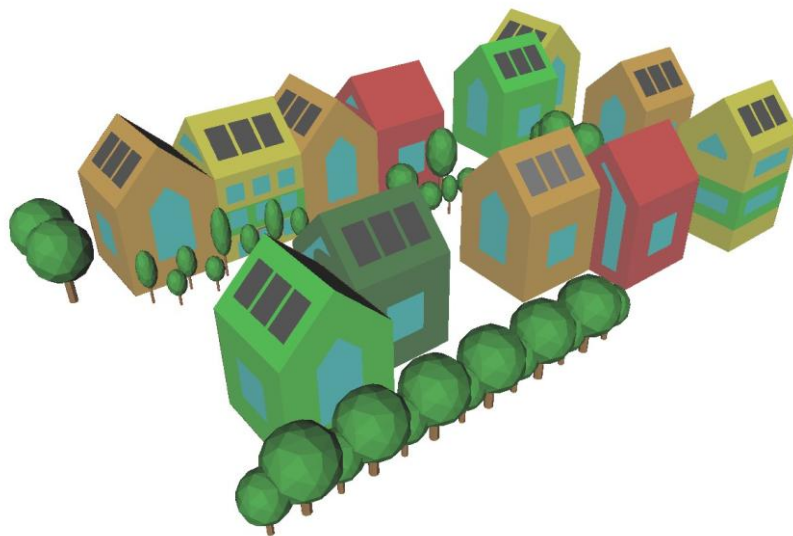




Beta 8

UML diagrams



Last updated on 14 May 2026





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The latest version of this document can be downloaded from:

https://github.com/tudelft3d/Energy_ADE/tree/3.0beta8/documentation

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<https://3d.bk.tudelft.nl/urbanrg/>

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Driving Urban
Transitions



Co-funded by
the European Union



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Acknowledgements

The author would like to thank the following people for their suggestions and fruitful discussions that have helped design the current version of the Energy ADE conceptual data model:

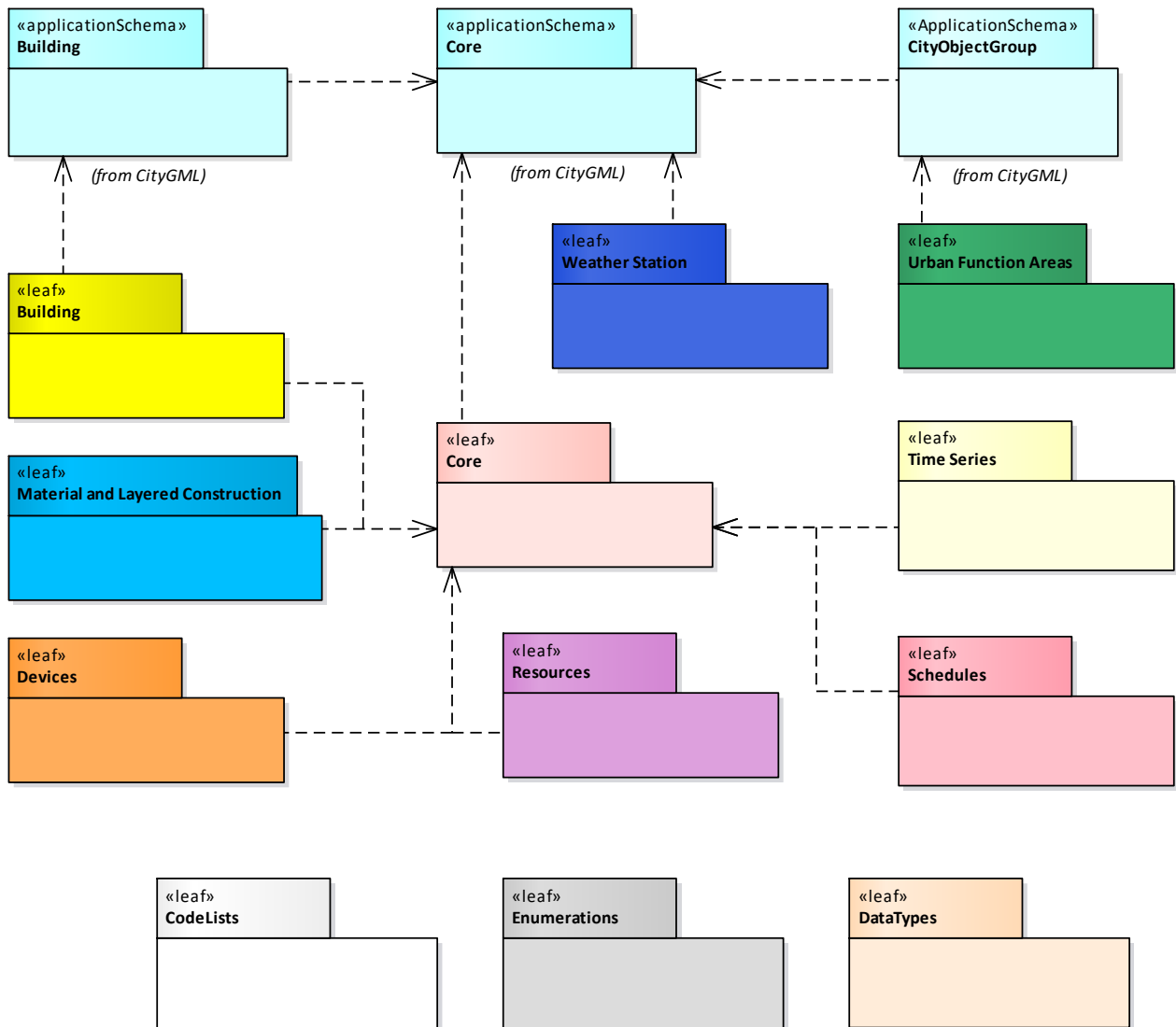
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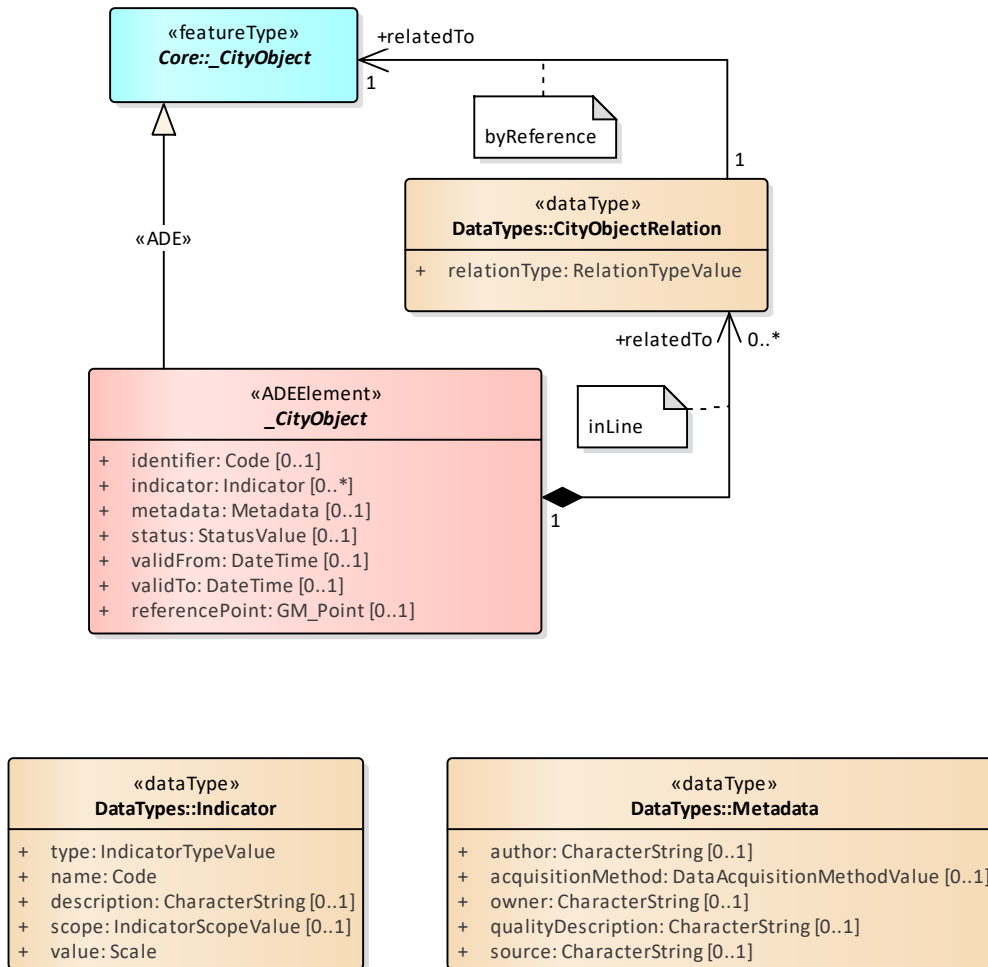
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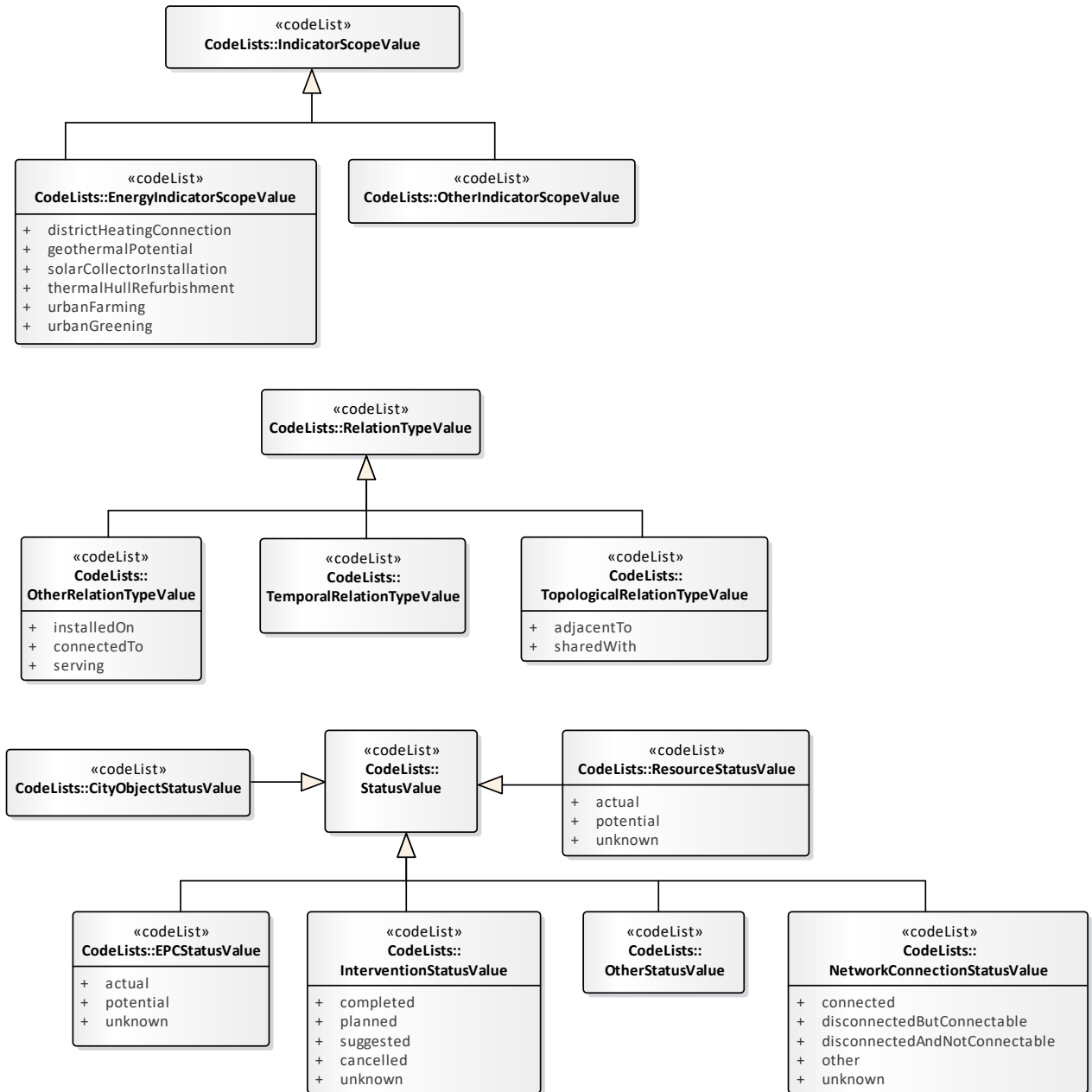
Overview of the packages



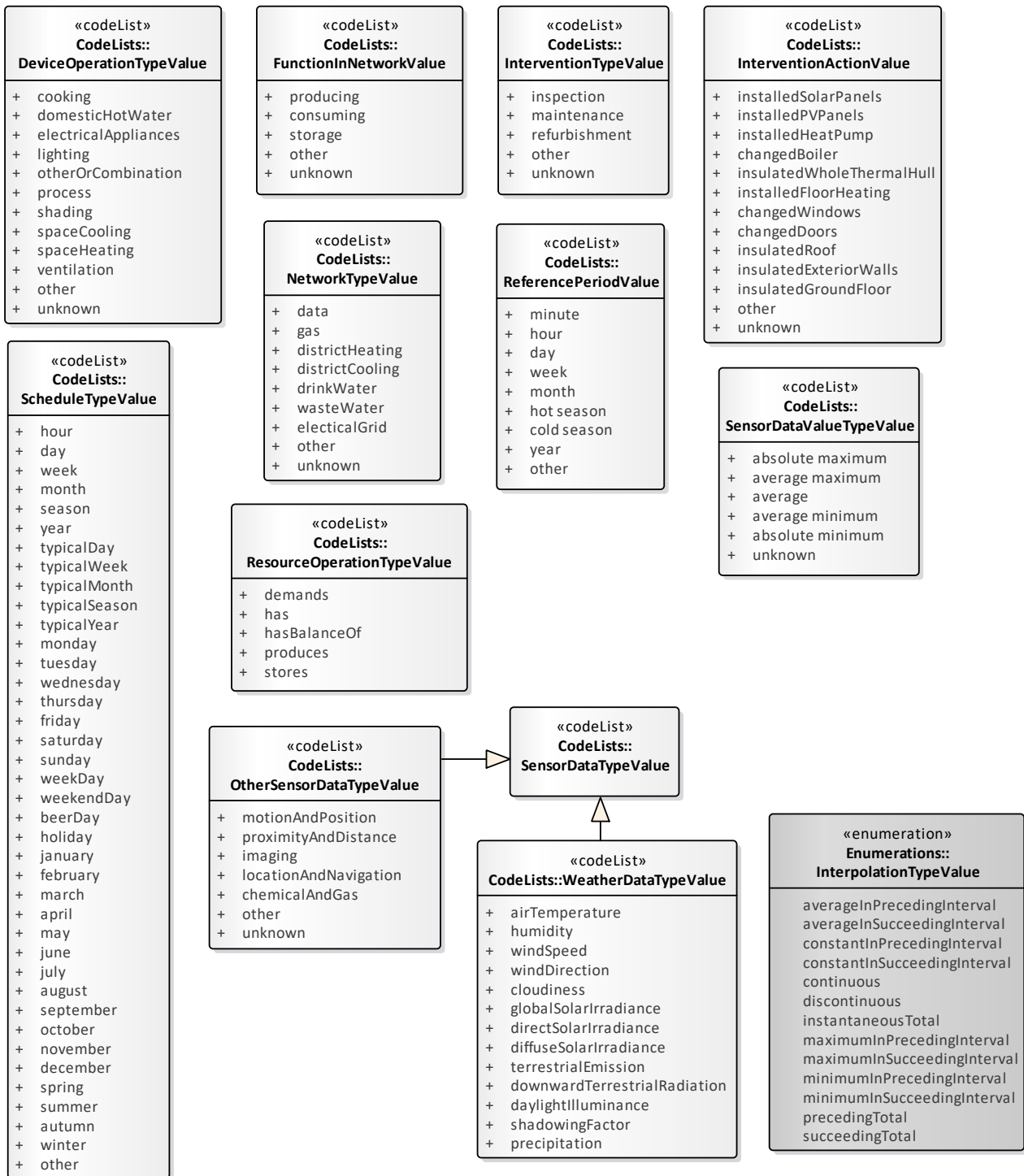
Core package part 1



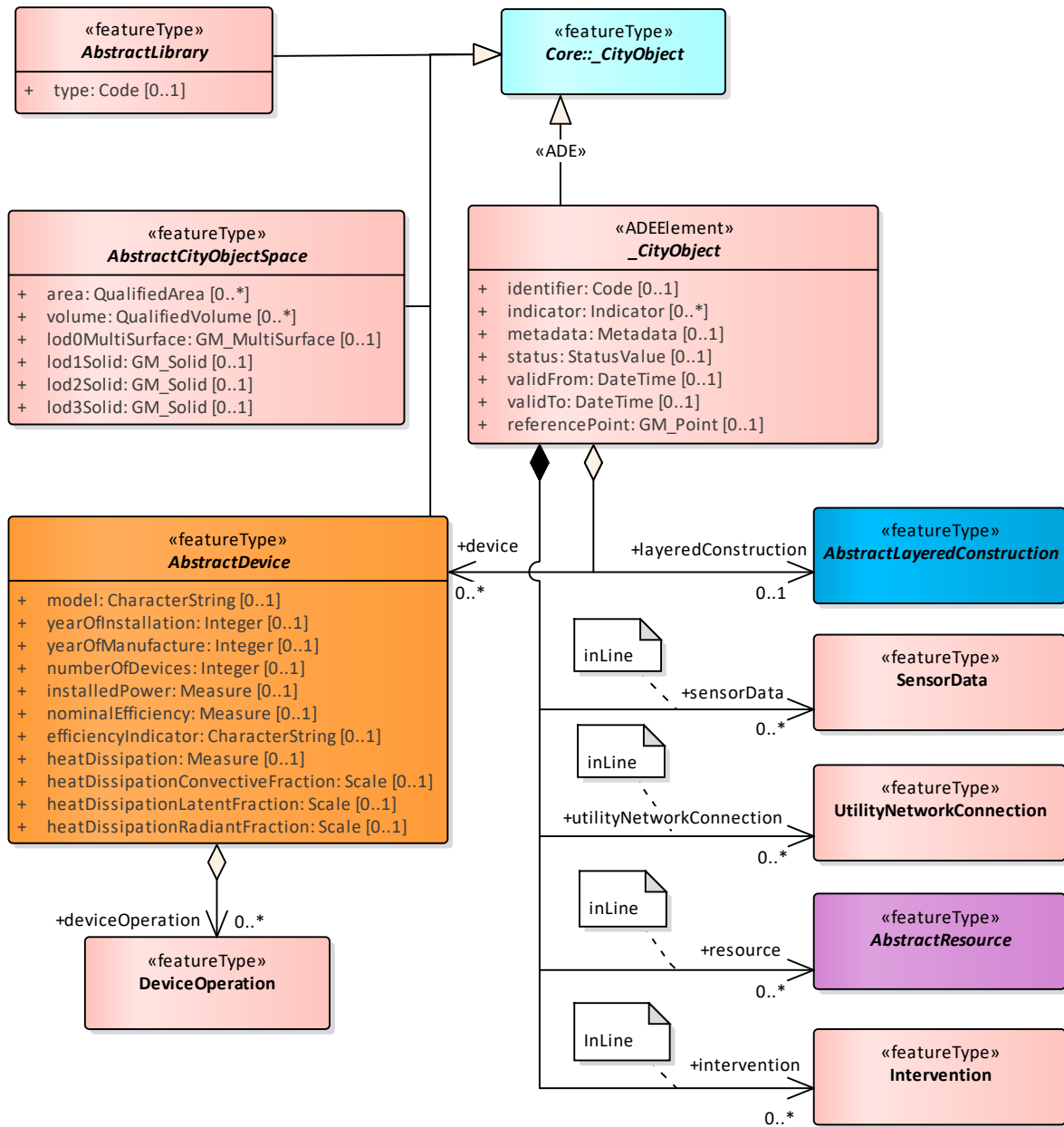
Core package part 1: CodeLists



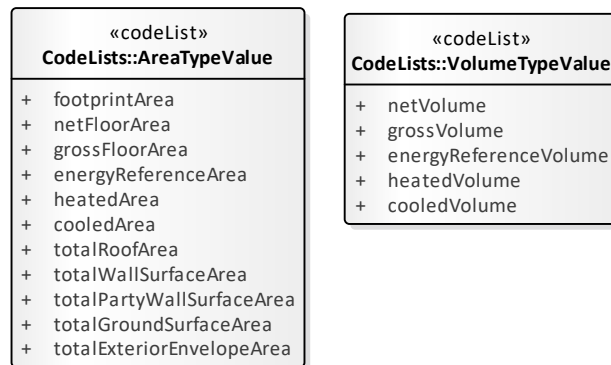
Core package part 2: CodeLists and Enumerations



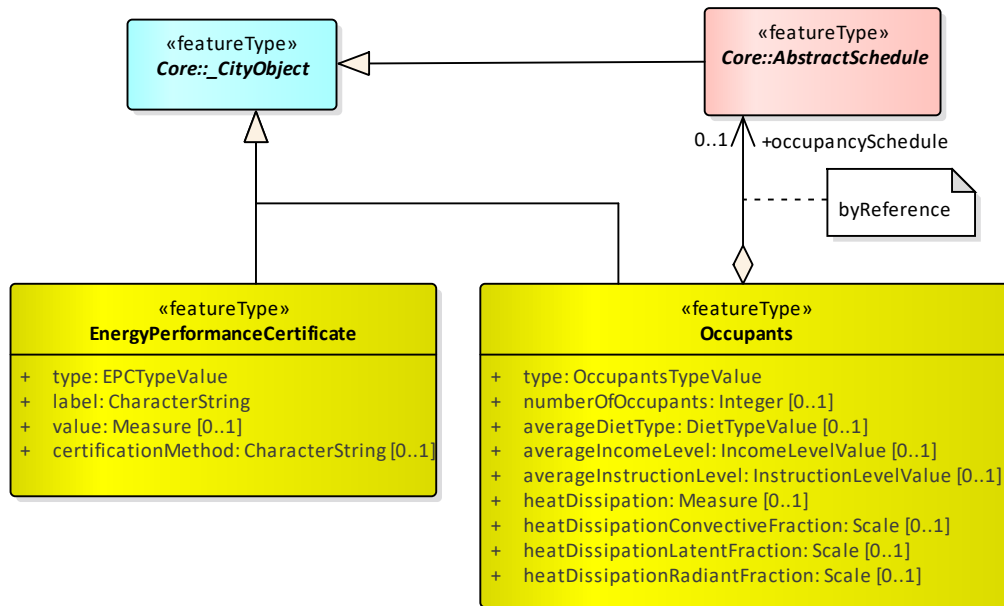
Core package part 3

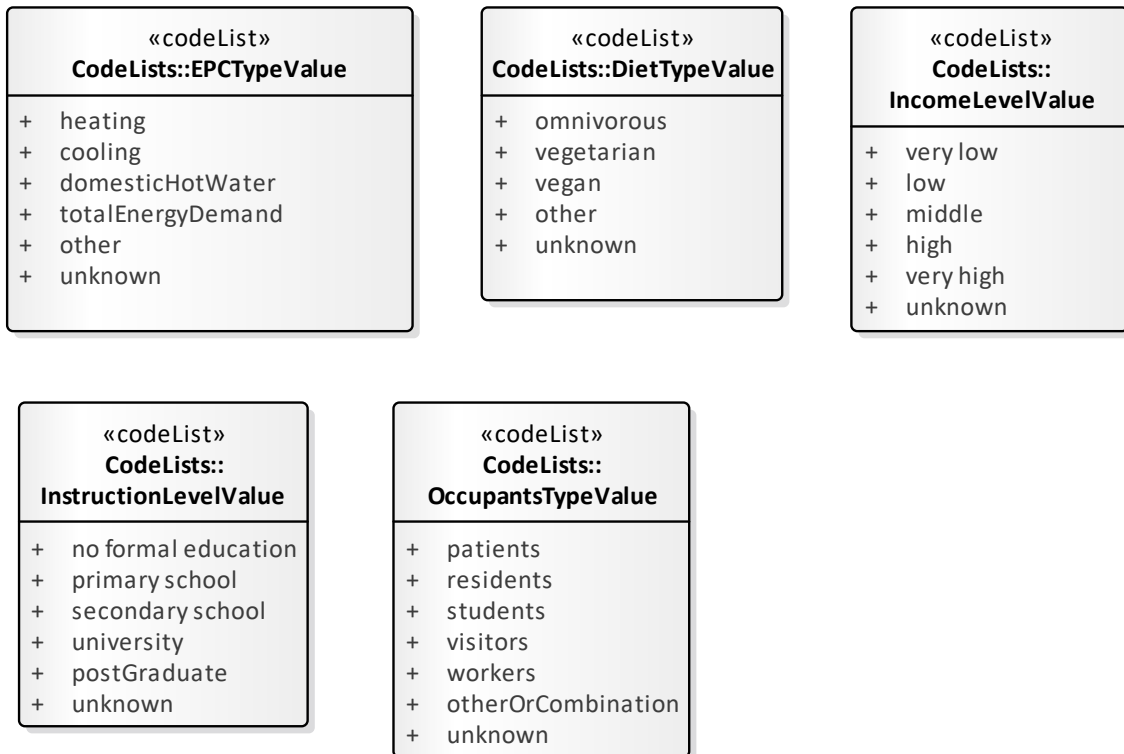


Core package part 3: CodeLists

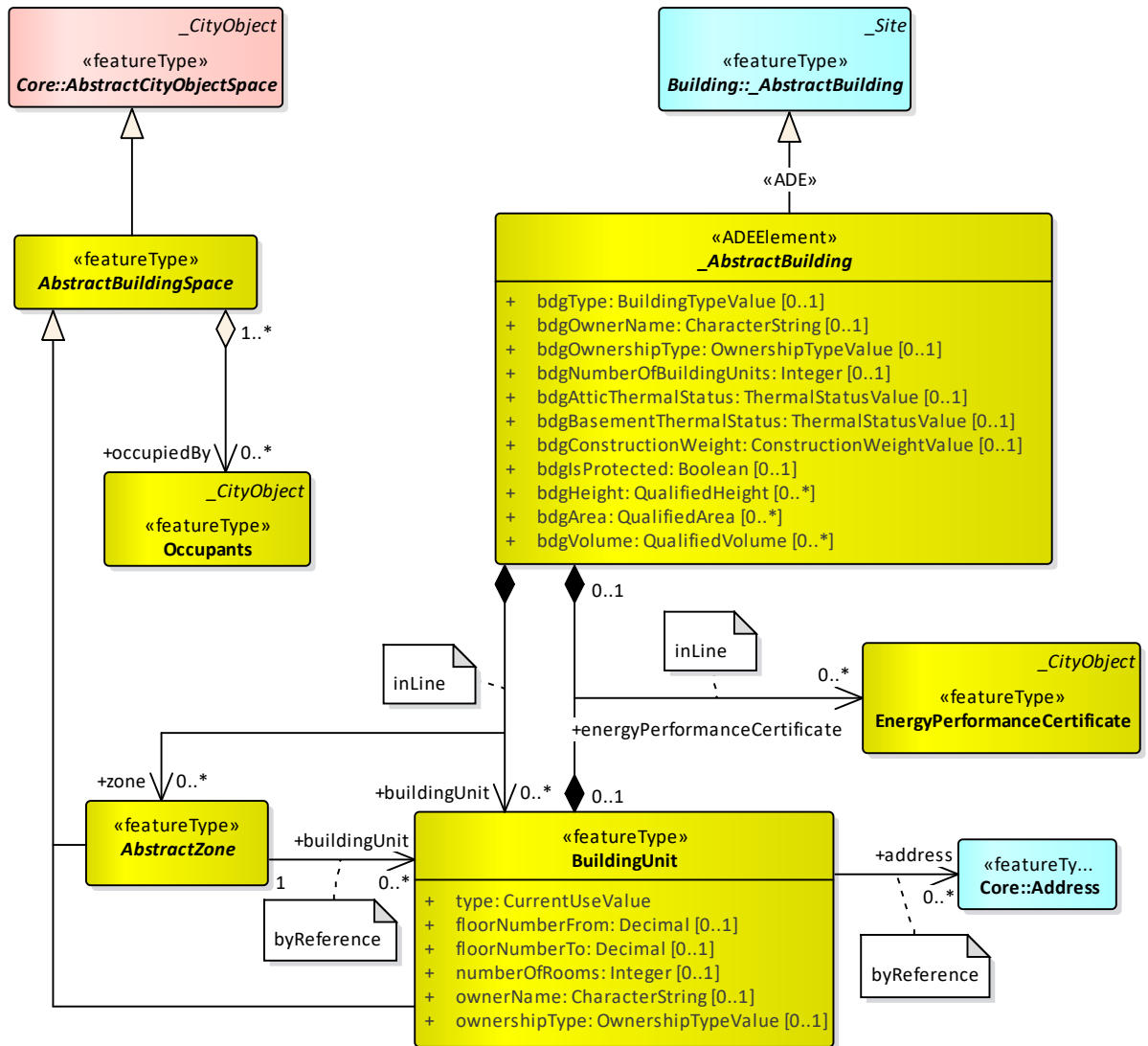


Building package part 1

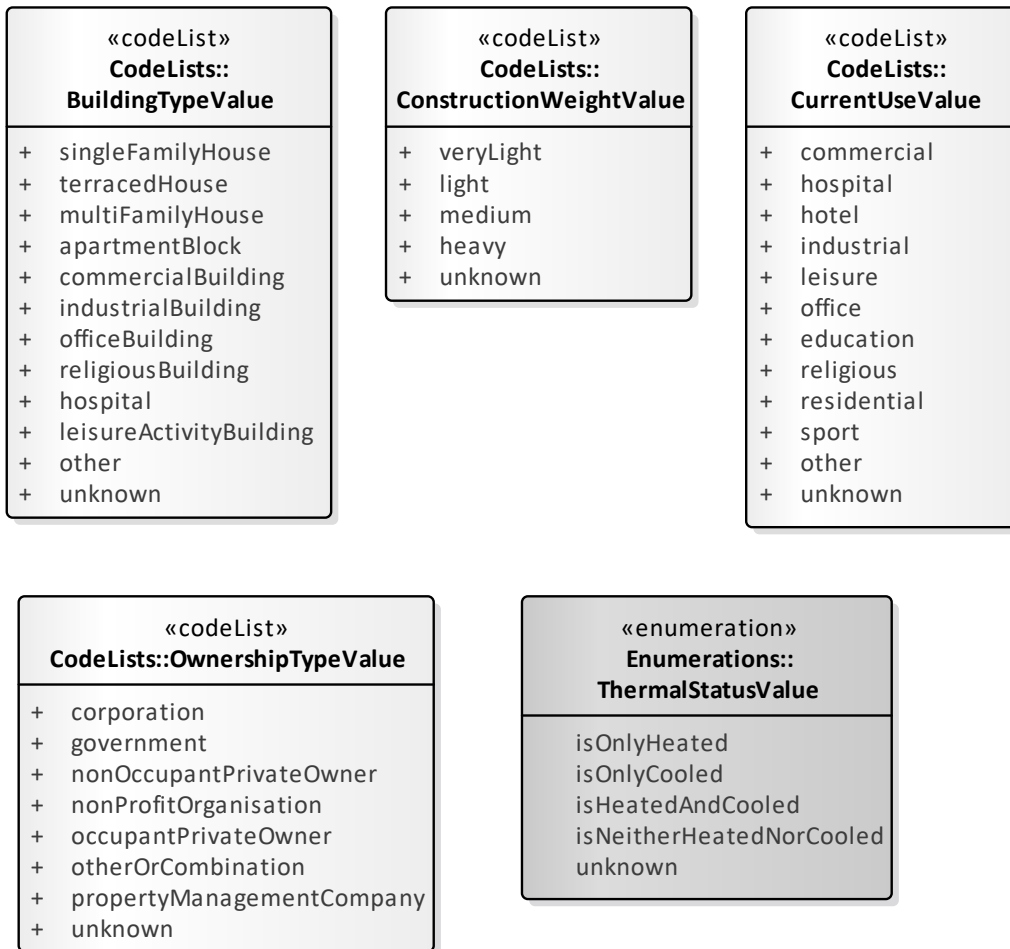


Building package part 1: CodeLists and Enumerations

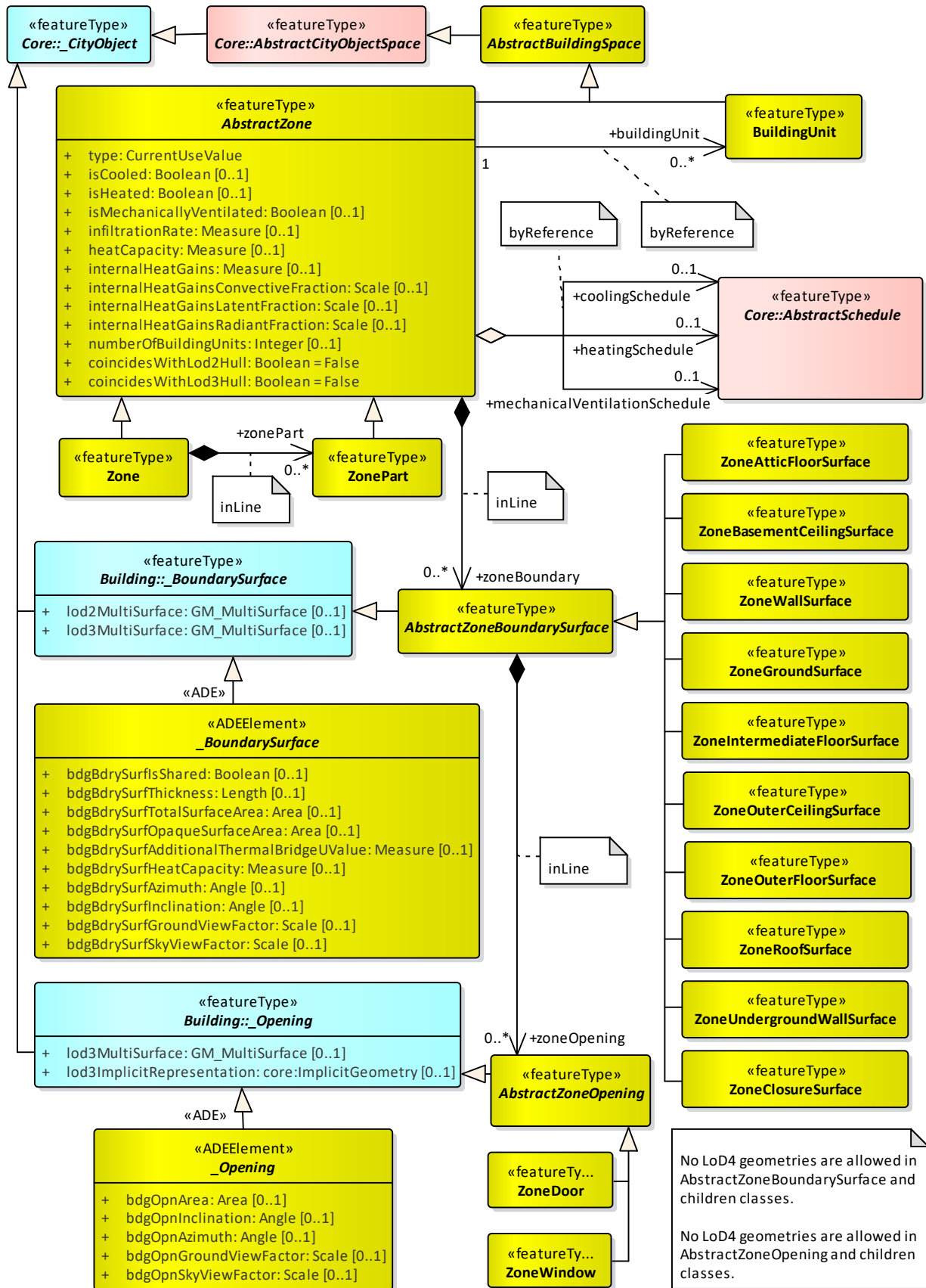
Building package part 2



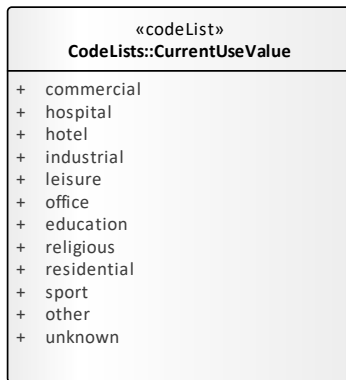
Building package part 2: CodeLists and Enumerations



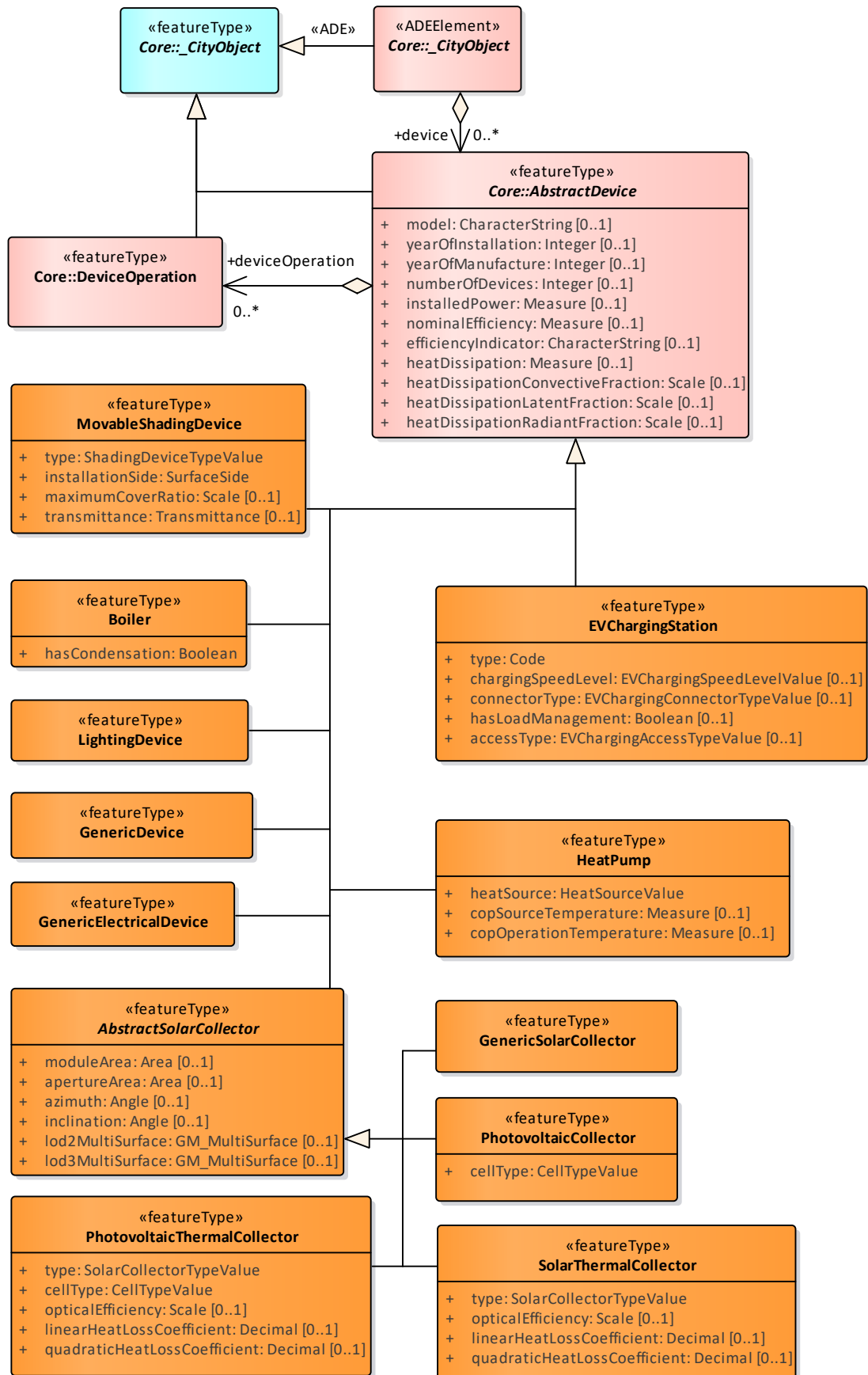
Building package part 3



Building package part 3: CodeLists and Enumerations



Devices package part 1



Devices package part 1: CodeLists and Enumerations

«codeList» CodeLists::CellTypeValue
+ monocrystalline + polycrystalline + amorphous + other + unknown

«codeList» CodeLists::EVChargingAccessValue
+ private + public + unknown

«codeList» CodeLists::EVChargingConnectorTypeValue
+ AC - J1772 (Type1) + AC - Mennekes (Type2) + AC - GB/T + AC/DC - Tesla + DC - CHAdeMO + DC - CCS1 + DC - CCS2 + DC - GB/T + unknown

«codeList» CodeLists::EVChargingSpeedLevelValue
+ Level 1 + Level 2 + Level 3

«codeList» CodeLists::EVChargingStationTypeValue
+ AC + DC + unknown

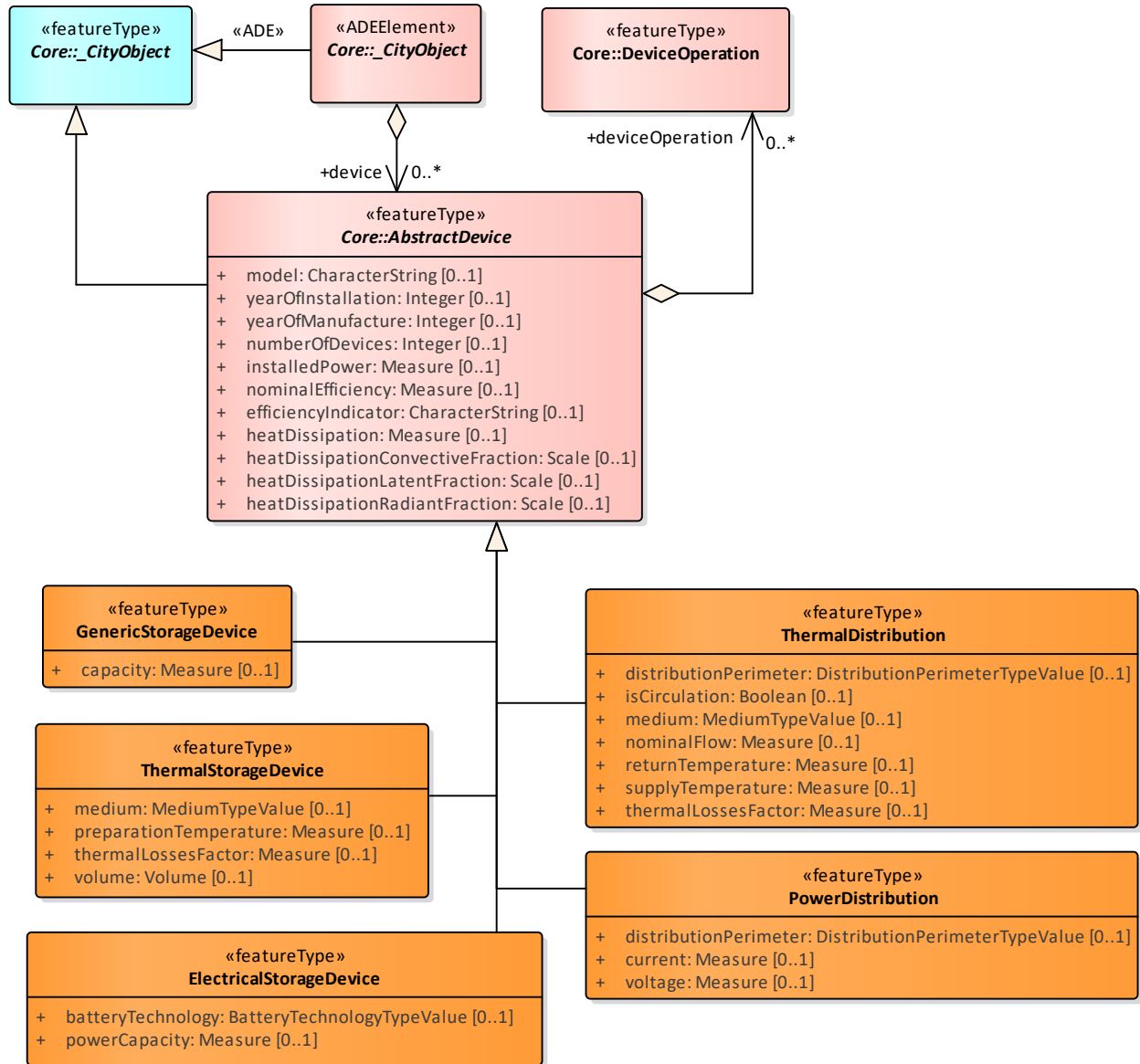
«codeList» CodeLists::HeatSourceValue
+ ambientAir + aquifer + exhaustAir + horizontalGroundCollector + verticalGroundCollector + unknown

«codeList» CodeLists::ShadingDeviceTypeValue
+ curtain + shutter + blind + other + unknown

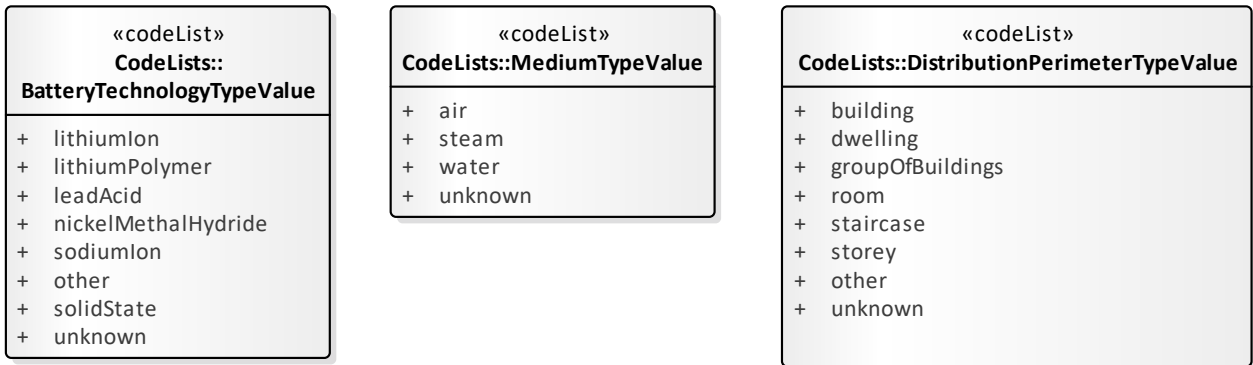
«codeList» CodeLists::SolarCollectorTypeValue
+ flatPlaneCollector + evacuatedTubeCollector + unknown

«enumeration» Enumerations::SurfaceSide
inside outside

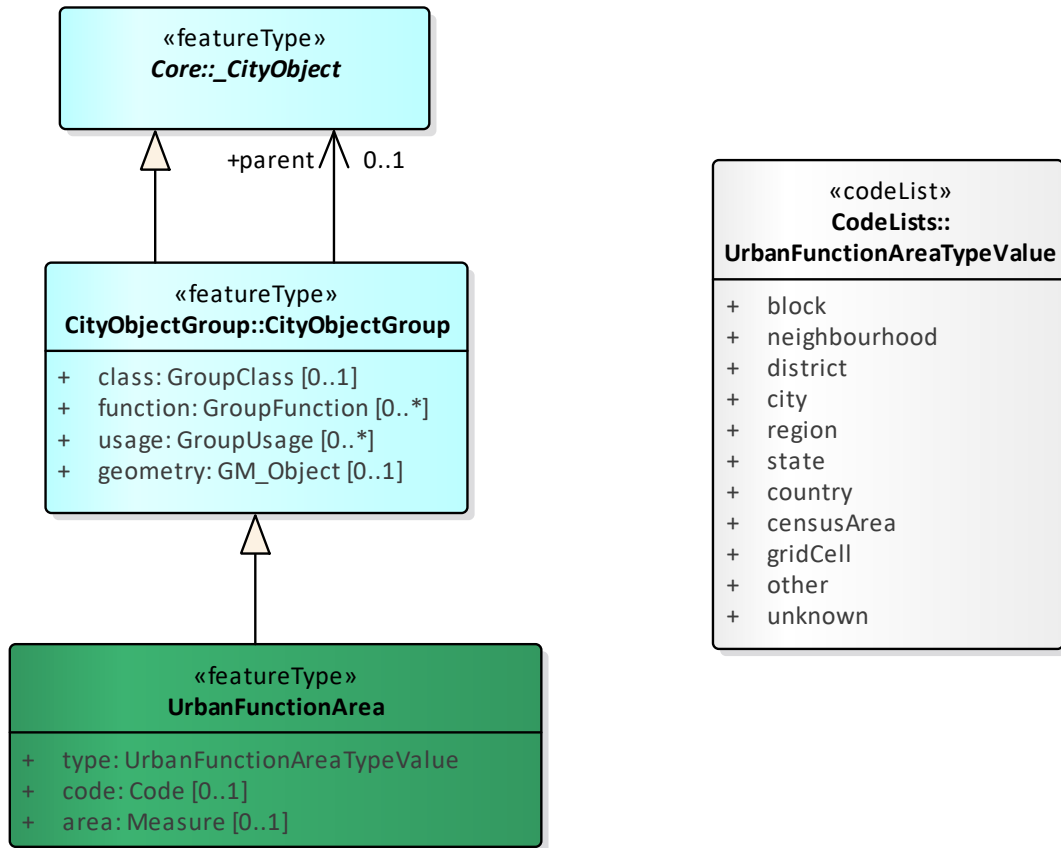
Devices package part 2



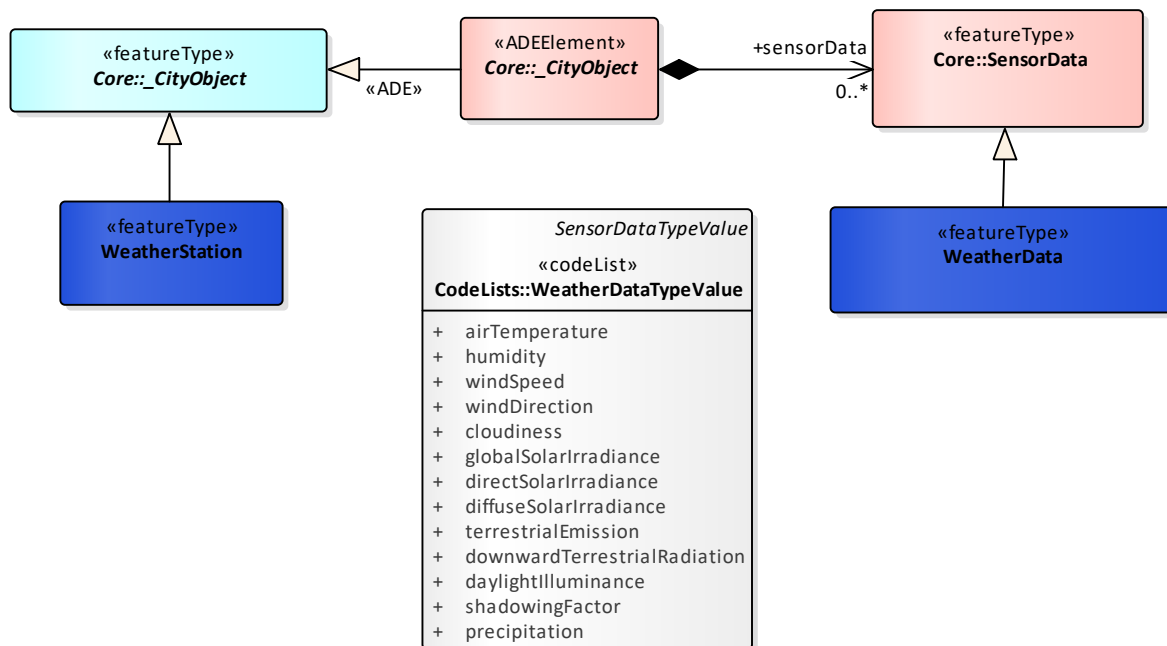
Devices package part 2: CodeLists and Enumerations



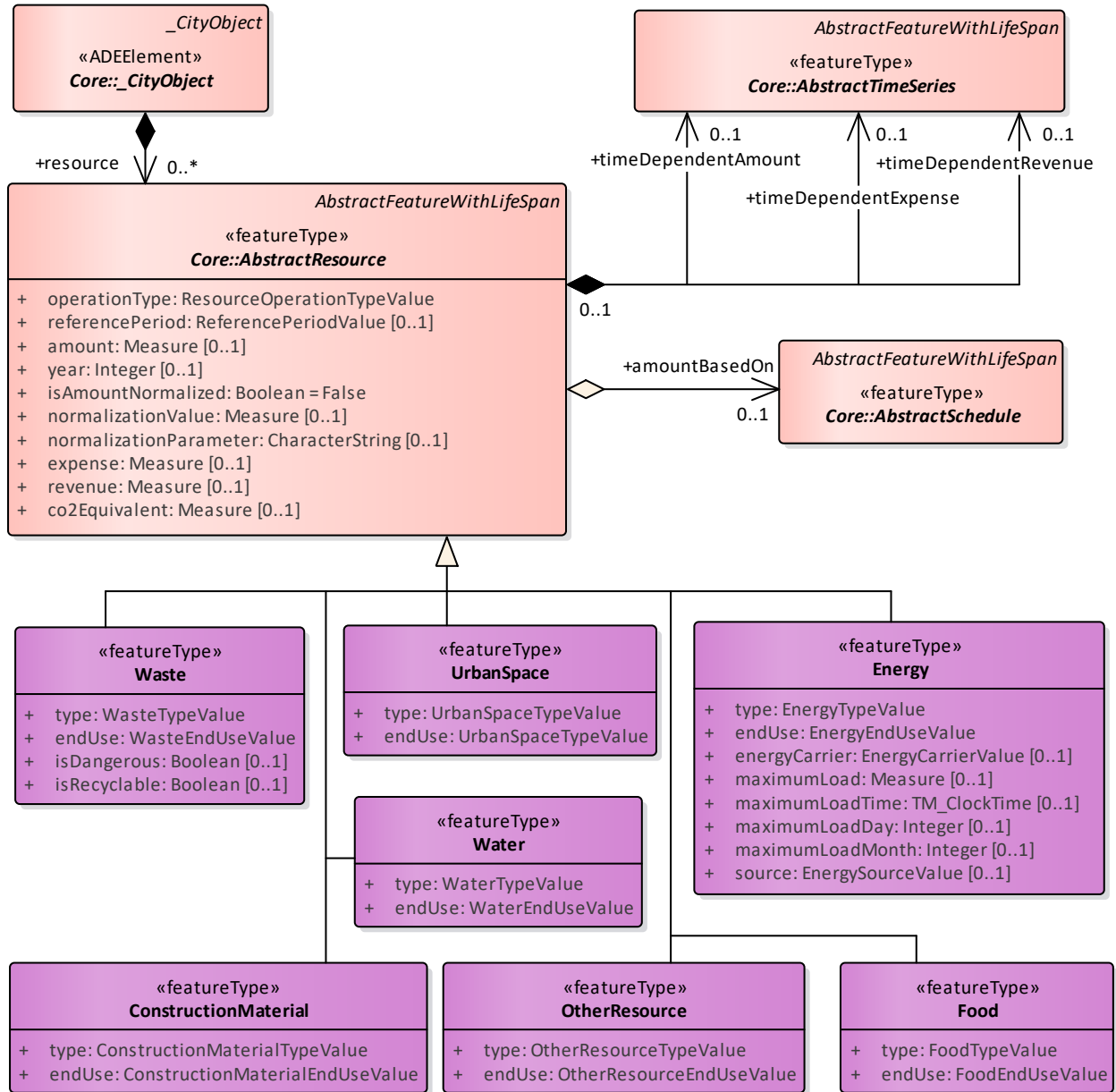
Urban function areas package and CodeLists



Weather station package and CodeLists



Resources package



Resources package: CodeLists and Enumerations

«codeList» CodeLists:: ConstructionMaterialEndUseValue
+ construction + refurbishment + otherOrCombination + unknown

«codeList» CodeLists:: ConstructionMaterialTypeValue
+ wood + tiles + concrete + sand + soil + otherOrCombination + glass + bricks + unknown

«codeList» CodeLists:: EnergyCarrierValue
+ electricity + coal + naturalGas + chilledAir + chilledWater + fuelOil + hotAir + steam + hotWater + woodChips + hydrogen + propane + woodPallets + other + unknown

«codeList» CodeLists:: EnergyEndUseValue
+ cooking + domesticHotWater + electricalAppliances + lighting + mobility + otherOrCombination + spaceCooling + spaceHeating + ventilation + process + construction + unknown

«codeList» CodeLists:: EnergyTypeValue
+ actual + net + primary + secondary + final + useful

«codeList» CodeLists:: EnergySourceValue
+ photovoltaicPanels + powerGrid + districtHeating + solarThermalPanels + heatPump + wind

«codeList» CodeLists:: FoodEndUseValue
+ alimentation + commerce

«codeList» CodeLists:: FoodTypeValue
+ greeneries + meat + cereals + cheese + milk + wine + beer + unknown

«codeList» CodeLists:: OtherResourceEndUseValue

«codeList» CodeLists:: OtherResourceTypeValue

«codeList» CodeLists:: UrbanSpaceEndUseValue
--

«codeList» CodeLists:: UrbanSpaceTypeValue
--

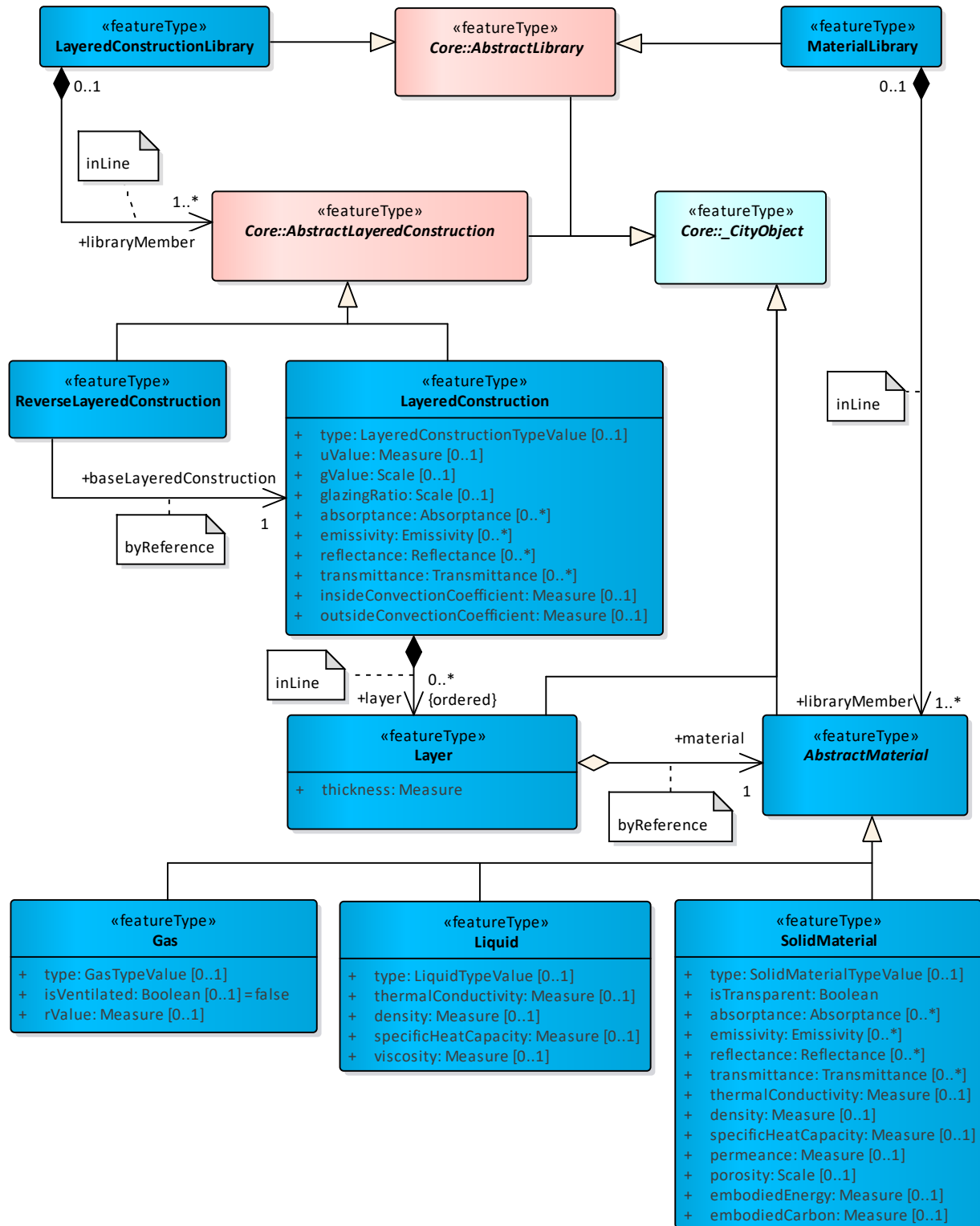
«codeList» CodeLists:: WasteEndUseValue
+ incineration + landFilling + recycling + other + unknown

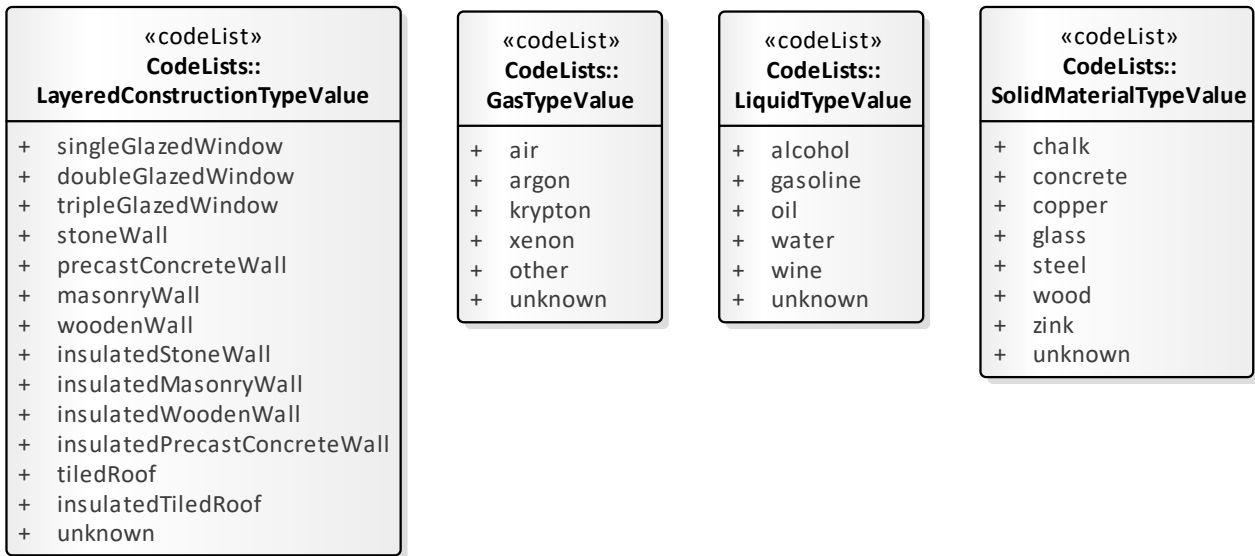
«codeList» CodeLists:: WasteTypeValue
+ organic + plastic + glass + paper + metal + water + otherOrCombination + unknown

«codeList» CodeLists:: WaterEndUseValue
+ drinking + washing + processing + otherOrCombination + unknown

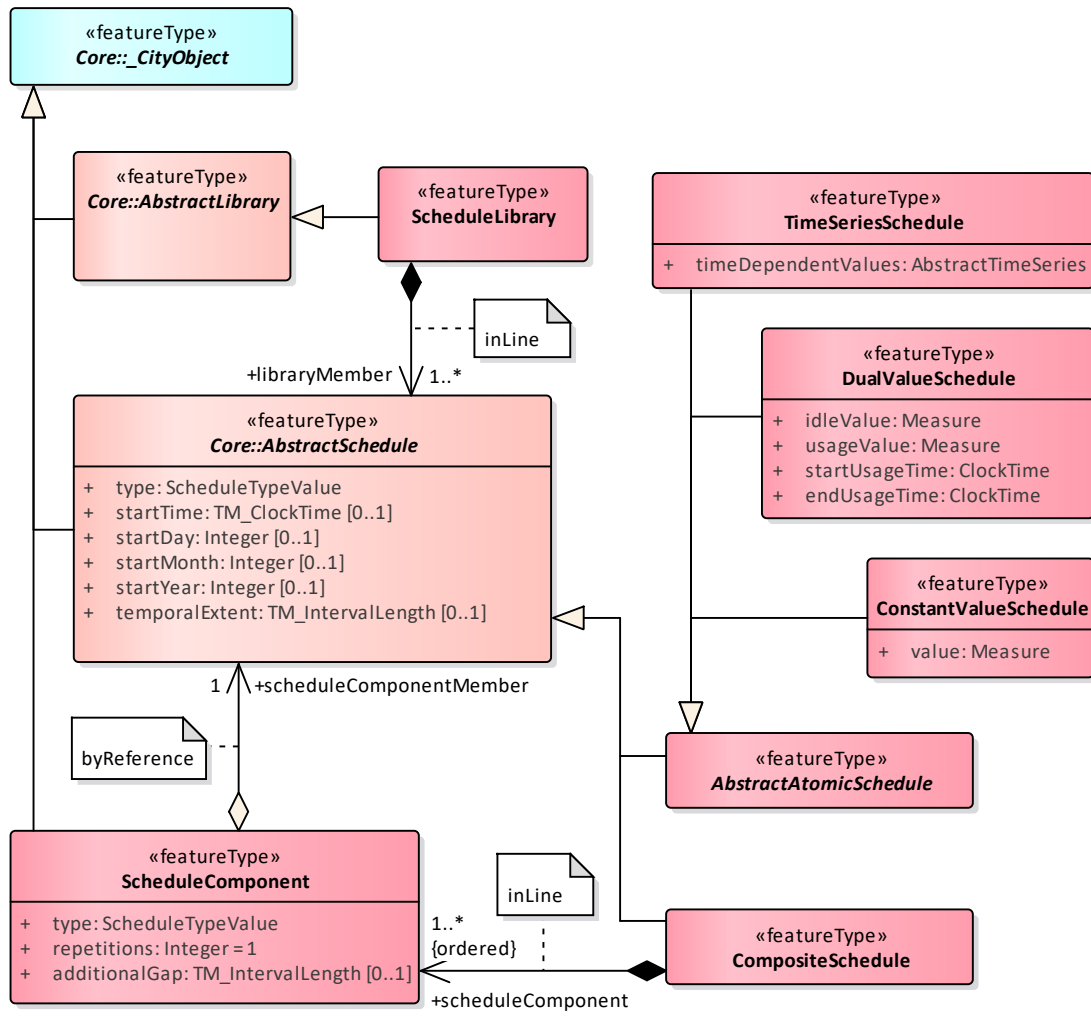
«codeList» CodeLists:: WaterTypeValue
+ drinkWater + wasteWater + rainWater + otherOrCombination + unknown

Layered construction package



Layered construction package: CodeLists

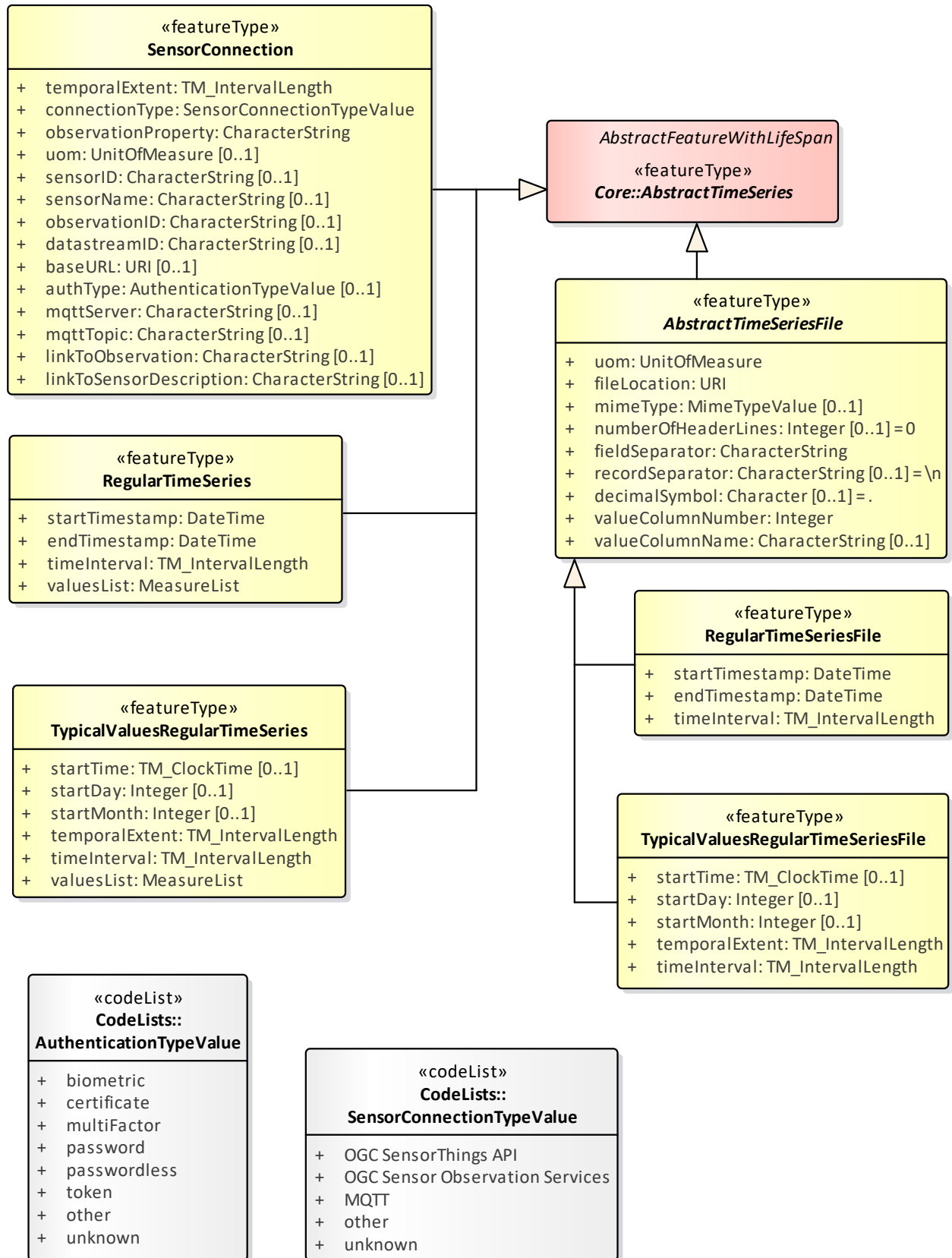
Schedules package



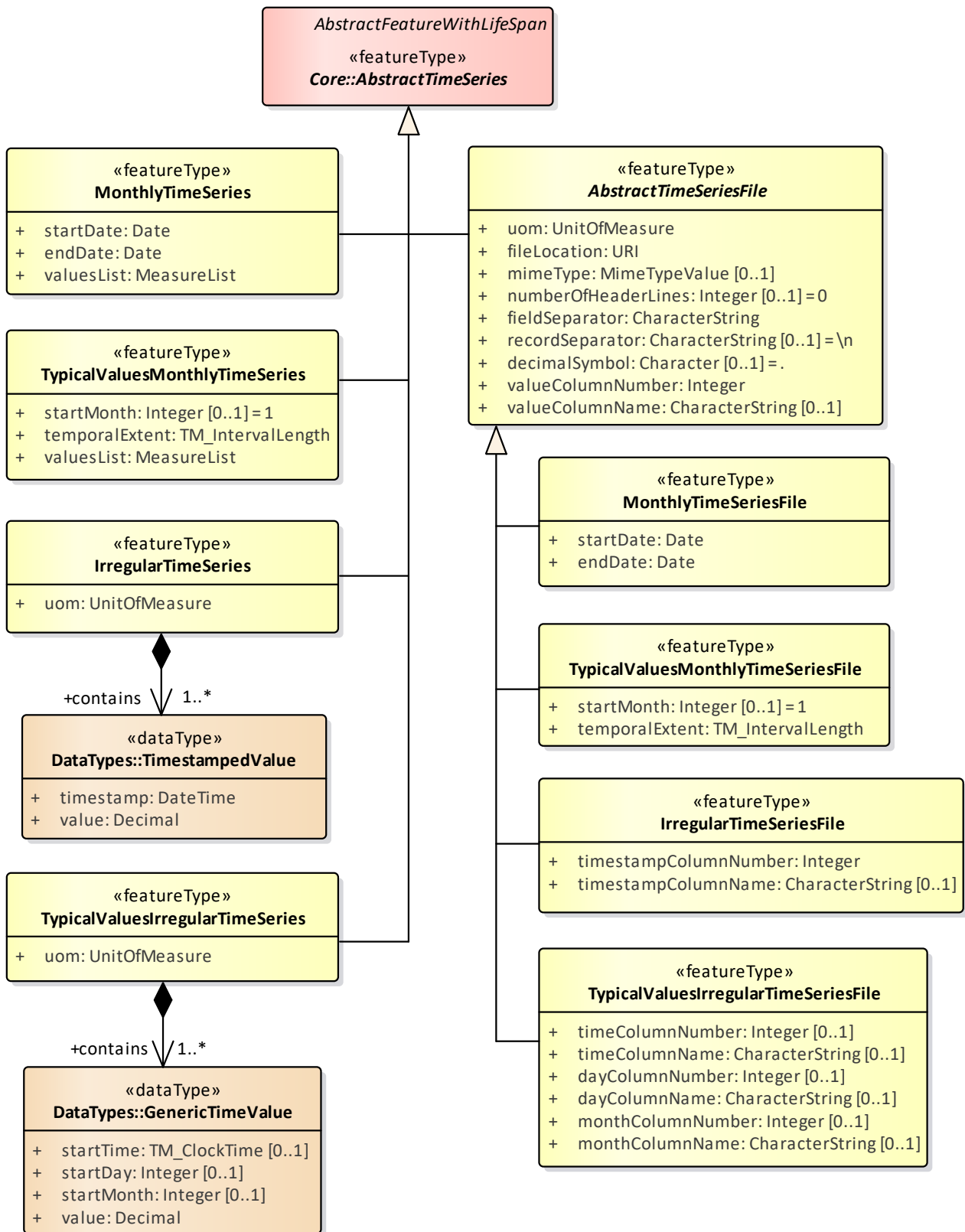
Schedules package: CodeLists

«codeList» CodeLists:: ScheduleTypeValue	
+	hour
+	day
+	week
+	month
+	season
+	year
+	typicalDay
+	typicalWeek
+	typicalMonth
+	typicalSeason
+	typicalYear
+	monday
+	tuesday
+	wednesday
+	thursday
+	friday
+	saturday
+	sunday
+	weekDay
+	weekendDay
+	beerDay
+	holiday
+	january
+	february
+	march
+	april
+	may
+	june
+	july
+	august
+	september
+	october
+	november
+	december
+	spring
+	summer
+	autumn
+	winter
+	other

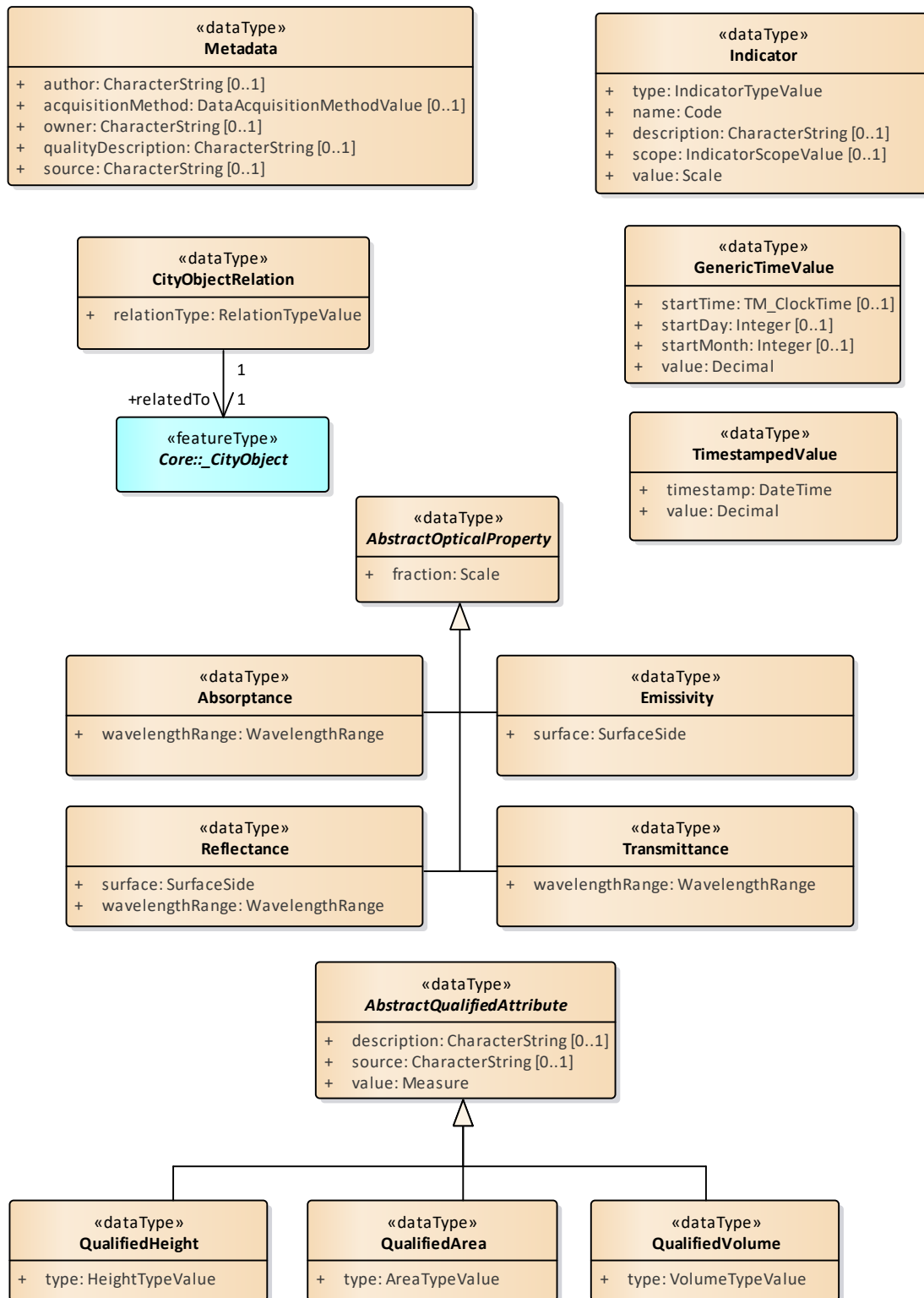
Time series package part 1 and CodeLists



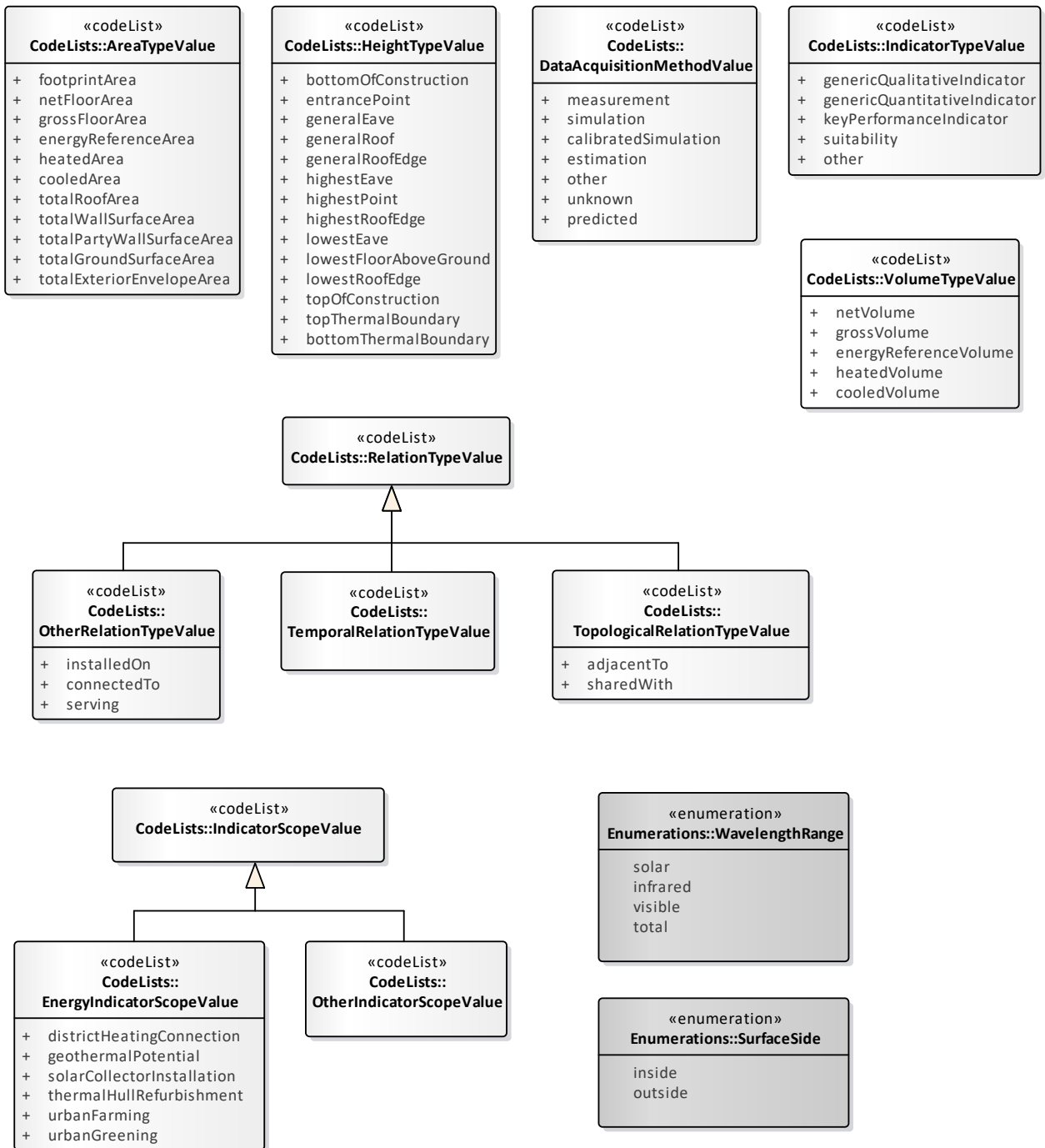
Time series package part 2



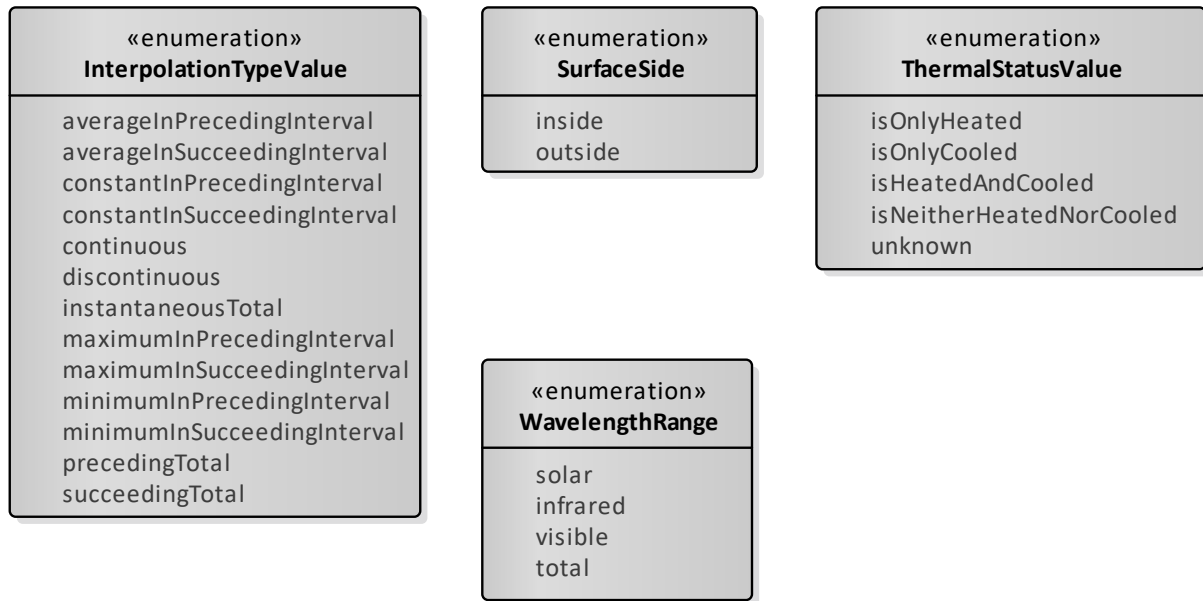
DataTypes package



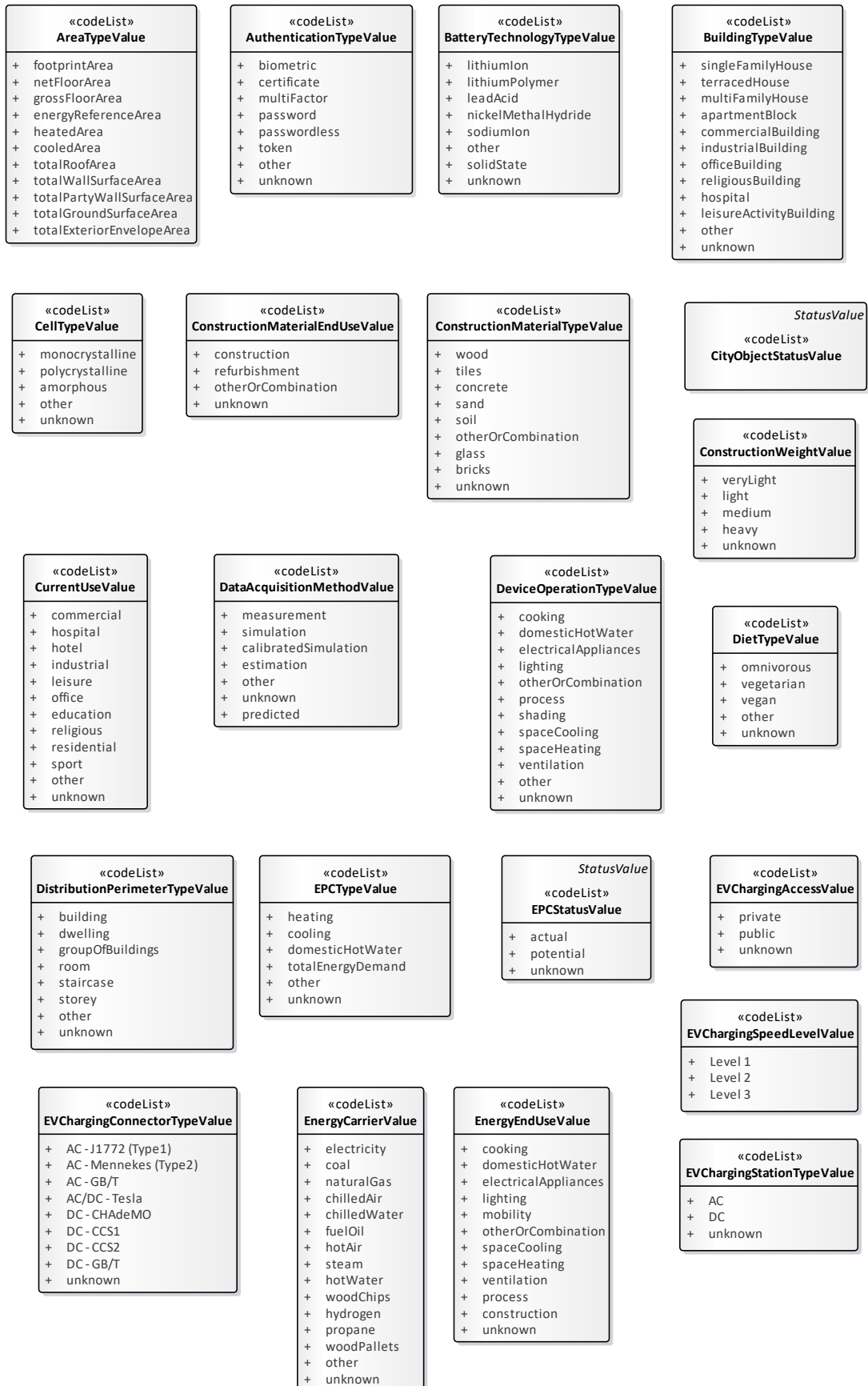
DataTypes package: CodeLists and Enumerations



Enumerations package



Codelists package



«codeList» EnergyIndicatorScopeValue
+ districtHeatingConnection
+ geothermalPotential
+ solarCollectorInstallation
+ thermalHullRefurbishment
+ urbanFarming
+ urbanGreening

«codeList» EnergySourceValue
+ photovoltaicPanels
+ powerGrid
+ districtHeating
+ solarThermalPanels
+ heatPump
+ wind

«codeList» EnergyTypeValue
+ actual
+ net
+ primary
+ secondary
+ final
+ useful

«codeList» FoodEndUseValue
+ alimentation
+ commerce

«codeList» FoodTypeValue
+ greeneries
+ meat
+ cereals
+ cheese
+ milk
+ wine
+ beer
+ unknown

«codeList» FunctionInNetworkValue
+ producing
+ consuming
+ storage
+ other
+ unknown

«codeList» GasTypeValue
+ air
+ argon
+ krypton
+ xenon
+ other
+ unknown

«codeList» HeatSourceValue
+ ambientAir
+ aquifer
+ exhaustAir
+ horizontalGroundCollector
+ verticalGroundCollector
+ unknown

«codeList» HeightTypeValue
+ bottomOfConstruction
+ entrancePoint
+ generalEave
+ generalRoof
+ generalRoofEdge
+ highestEave
+ highestPoint
+ highestRoofEdge
+ lowestEave
+ lowestFloorAboveGround
+ lowestRoofEdge
+ topOfConstruction
+ topThermalBoundary
+ bottomThermalBoundary

«codeList» IndicatorTypeValue
+ genericQualitativeIndicator
+ genericQuantitativeIndicator
+ keyPerformanceIndicator
+ suitability
+ other

«codeList» IndicatorScopeValue

«codeList» IncomeLevelValue
+ very low
+ low
+ middle
+ high
+ very high
+ unknown

«codeList» InstructionLevelValue
+ no formal education
+ primary school
+ secondary school
+ university
+ postGraduate
+ unknown

«codeList» InterventionActionValue
+ installedSolarPanels
+ installedPVPanels
+ installedHeatPump
+ changedBoiler
+ insulatedWholeThermalHull
+ installedFloorHeating
+ changedWindows
+ changedDoors
+ insulatedRoof
+ insulatedExteriorWalls
+ insulatedGroundFloor
+ other
+ unknown

StatusValue «codeList» InterventionStatusValue
+ completed
+ planned
+ suggested
+ cancelled
+ unknown

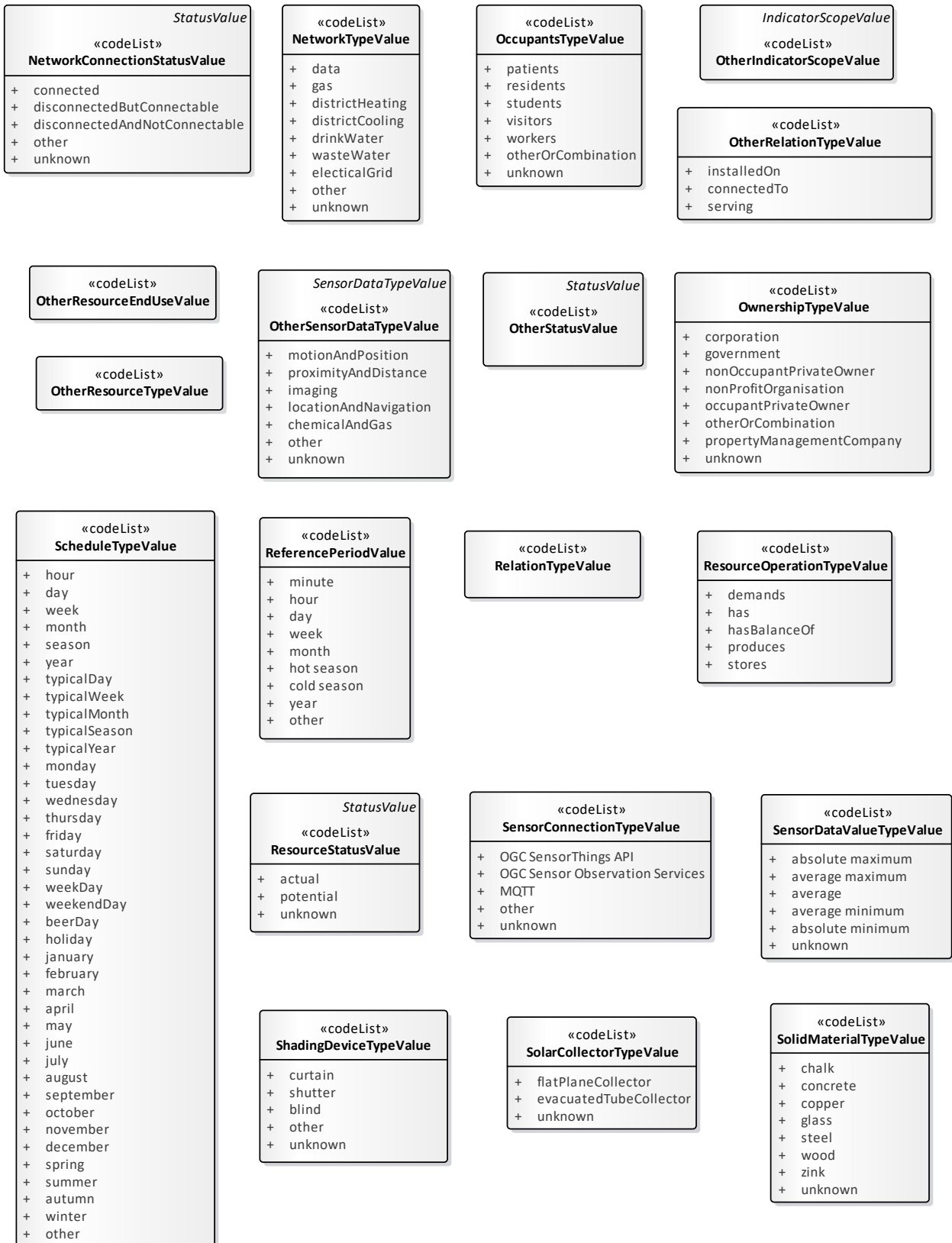
«codeList» InterventionTypeValue
+ inspection
+ maintenance
+ refurbishment
+ other
+ unknown

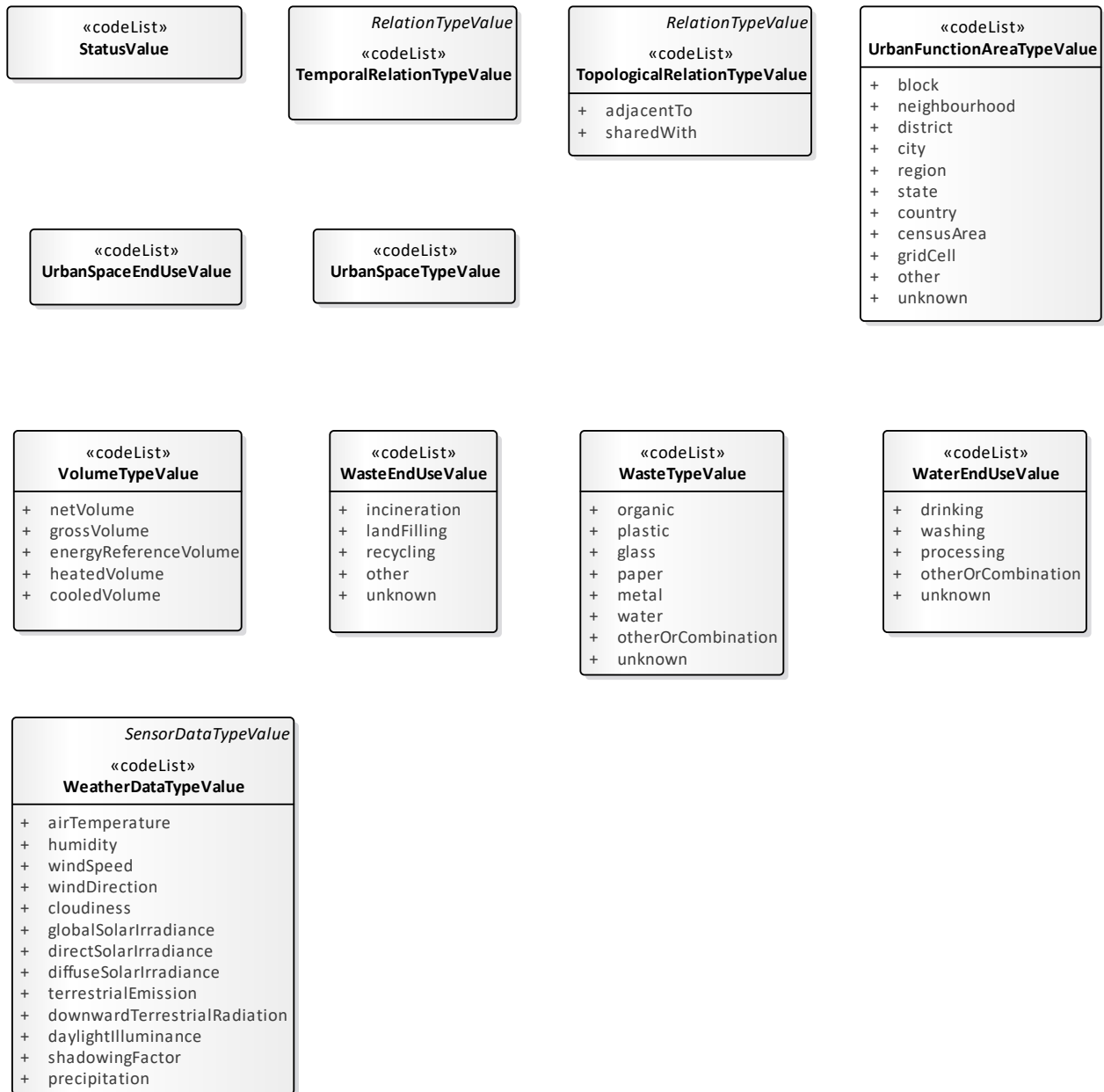
«codeList» LayeredConstructionTypeValue
+ singleGlazedWindow
+ doubleGlazedWindow
+ tripleGlazedWindow
+ stoneWall
+ precastConcreteWall
+ masonryWall
+ woodenWall
+ insulatedStoneWall
+ insulatedMasonryWall
+ insulatedWoodenWall
+ insulatedPrecastConcreteWall
+ tiledRoof
+ insulatedTiledRoof
+ unknown

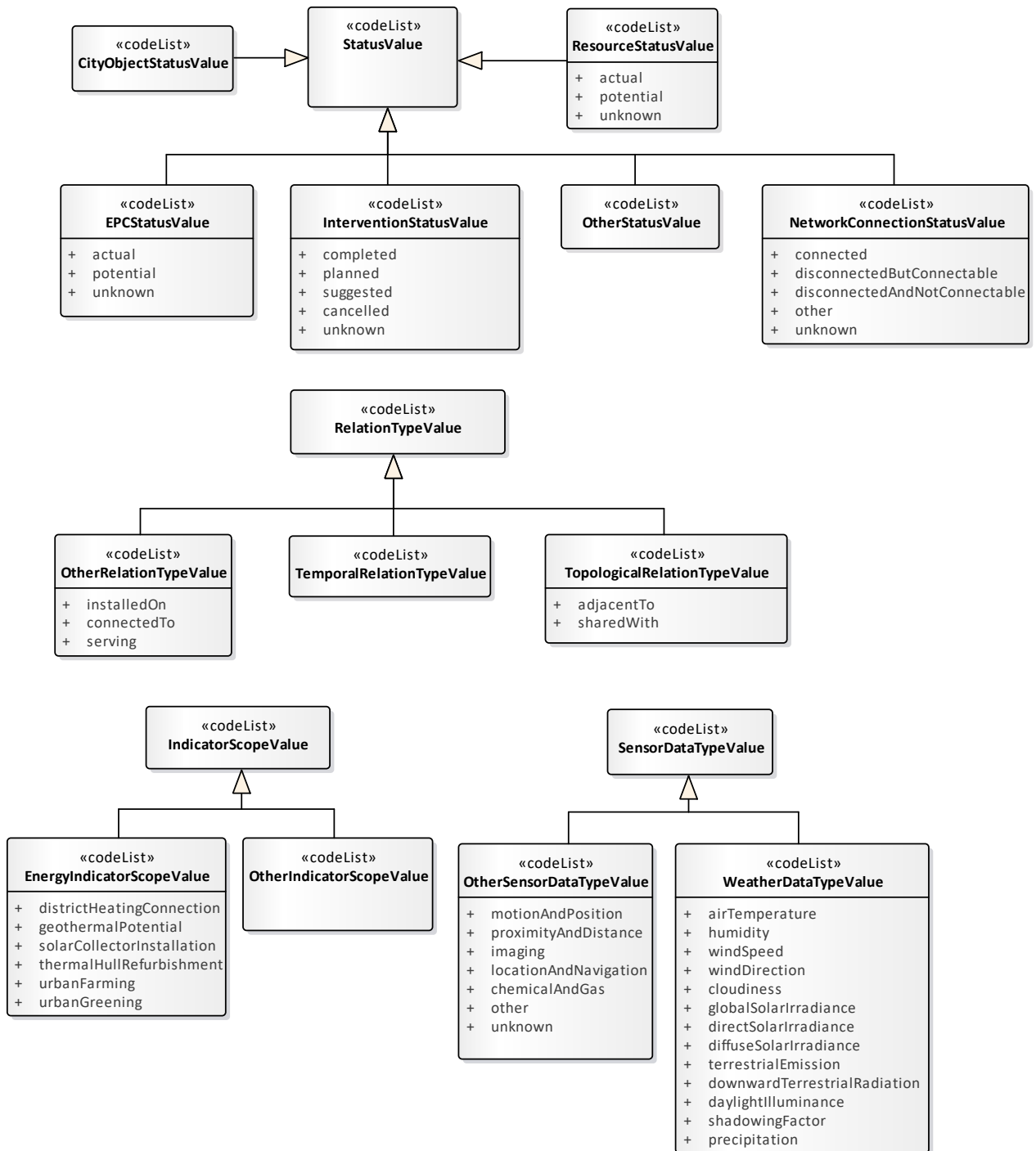
«codeList» LiquidTypeValue
+ alcohol
+ gasoline
+ oil
+ water
+ wine
+ unknown

«codeList» MediumTypeValue
+ air
+ steam
+ water
+ unknown

«codeList» MimeTypeValue







Changelog

Version 3.0 beta 8

Cumulative list of changes since beta 7 (as of 2026-05-14)

CodeLists package:

Several CodeList classes added and some renamed. Added values to several CodeLists.

Enumerations package:

- Class **NetworkConnectionStatusValue** has been changed to a CodeList and moved to the corresponding package.

DataTypes package:

- Dropped class **Suitability**
- Added class **Absorptance**
- Added class **CityObjectRelation**
- Added class **FeatureRelation**
- Added class **GenericTimeValue**
- Added class **Indicator**
- Added class **Metadata**
- Added class **TimestampedValue**

Core package:

- Dropped class **AbstractBuildingUnit**
- Dropped class **AbstractUsageZone**
- Dropped class **AbstractThermalZone**
- Dropped class **RefurbishmentMeasure**
- Moved class **EnergyPerformanceCertification** to Building package
- Moved class **ADE _AbstractBuilding** to Building package
- Renamed class **AbstractBuildingPartition** to **AbstractCityObjectSpace**
- Renamed class **WeatherData** to **SensorData**
- Added class **AbstractADEFeature**
- Added class **AbstractFeatureWithLifeSpan**
- Added class **Intervention**
- Classes **AbstractResource**, **AbstractSchedule**, **AbstractLayeredConstruction**, **UtilityNetworkCnnection**, **SensorData**, **DeviceOperation** are now derived from class **AbstractFeatureWithLifeSpan**
- Class **ADE _CityObject**:
 - Dropped property *suitability*
 - Renamed property from *weatherData* to *sensorData*
 - Added property *metadata*
 - Added property *identifier*
 - Added property *validFrom*
 - Added property *validTo*
 - Added property *status*
 - Added property *indicator*
- Class **AbstractLibrary**:
 - Dropped property *source*
 - Dropped property *author*
- Class **AbstractCityObjectSpace**:
 - Added property *lod0MultiSurface*
- Class **AbstractSchedule**:
 - Dropped property *libraryCode*
- Class **AbstractResource**:
 - Dropped property *status*
 - Dropped property *amountType*
 - Renamed property *costsMoney* to *expense*
 - Renamed property *yieldsMoney* to *revenue*

- Added property *referencePeriod*
- Added property *amountBasedOn*
- Added property *timeDependentExpense*
- Added property *timeDependentRevenue*
- Class **AbstractTimeSeries**:
 - Dropped property *acquisitionMethod*
 - Dropped property *source*
- Class **UtilityNetworkConnection**:
 - Dropped property *connectionStatus*

Building physics package:

Dropped. Some classes reworked and moved to new module Building

Occupancy package:

Dropped. Some classes reworked and moved to new module Building

Building package:

This module is new. It reworks and merges the previous Building physics module and Occupancy module. Additionally, it incorporates also class *ADE _AbstractBuilding* and its subclasses.

- Class **ADE _AbstractBuilding**
 - Class is moved from Core module
 - Dropped property *usageZone*
 - Dropped property *thermalZone*
 - Added property *occupiedBy*
 - Added property *zone*
 - Added property *owner*
 - Added property *ownershipType*
 - Added property *numberOfBuildingUnits*
- Class **EnergyPerformanceCertificate**
 - Class is moved from Core module
 - Class is derived from class **AbstractADEFeatureWithLifeSpan**
 - Dropped property *type*
 - Dropped property *issueDate*
 - Dropped property *expirationDate*
 - Dropped property *certificationUri*
- Class **Occupants**
 - Class is taken over from former Occupancy module
 - Class is now derived from class **AbstractADEFeatureWithLifeSpan**
 - Renamed relation *occupancyRate* to *occupancySchedule*
- Added class **AbstractBuildingSpace**
- Class **BuildingUnit**
 - Class is taken over from former Occupancy module
 - Class is now derived from class **AbstractBuildingSpace**
 - Added property *floorNumberFrom*
 - Added property *floorNumberTo*
- Added class **AbstractZone**
- Added class **Zone**
- Added class **ZonePart**
- Class **ADE _BoundarySurface**
 - Class is taken over from former Building physics module
 - Dropped property *bdgBdrySurfIsAdiabatic*
 - Added property *bdgBdrySurfIsShared*
 - Added property *bdgBdryAdditionalThermalBridgeUValue*
- Added class **AbstractZoneBoundarySurface**
- Added class **ZoneAtticFloorSurface**
- Added class **ZoneBasementCeilingSurface**
- Added class **ZoneExteriorWallSurface**
- Added class **ZoneGroundSurfaceSurface**

- Added class **ZoneIntermediateFloorSurface**
- Added class **ZoneRoofSurface**
- Added class **ZoneUndergrounSurface**
- Added class **ZoneClosureSurface**
- Added class **ADE _Opening**
 - Class is taken over from former Building physics module
- Added class **ADE AbstractZoneOpening**
- Added class **ZoneDoor**
- Added class **ZoneWindow**

Devices package:

- Added class **GenericStorageDevice**
- Added class **EVChargingStation**
- Added class **ThermalDistribution**
- Added class **PowerDistribution**
- Class **AbstractDevice**
 - Added property *yearOfInstallation*

Urban function area package

- Class **UrbanFunctionArea**
 - Added property *area*
 - Set type of property *code* to gml:Code

WeatherStation package

- Added class **WeatherData**

Resources package

- Class **Energy**
 - Added property *maximulLoad*
 - Added property *maximulLoadTime*
 - Added property *maximulLoadDay*
 - Added property *maximulLoadMonth*

Layered Construction package

- Class **MaterialLibrary**
 - Changed multiplicity of property *libraryMember* to 1..*
- Class **LayeredConstructionLibrary**
 - Changed multiplicity of property *libraryMember* to 1..*
- Class **LayeredConstructionMaterial**
 - Dropped property *libraryCode*
 - Added property *type*
 - Added property *absorptance*
 - Added property *insideConvectionCoefficient*
 - Added property *outsideConvectionCoefficient*
- Class **AbstractMaterial**
 - Class is now derived from class **AbstractADEFeatureWithLifeSpan**
 - Dropped property *libraryCode*
- Class **Layer**
 - Class is now derived from class **AbstractADEFeatureWithLifeSpan**
- Class **SolidMaterial**
 - Added property *type*
 - Added property *isTransparent*
 - Added property *absorptance*
 - Added property *emissivity*
 - Added property *reflectance*
 - Added property *transmittance*
- Added class **Liquid**
- Class **Gas**
 - Added property *type*

Schedules package:

- Dropped class **AtomicSchedule**
- Dropped class **AtomicScheduleValue**
- Added class **AbstractAtomicSchedule**
- Added class **ConstantValueSchedule**
- Added class **TimeSeriesSchedule**
- Class **DualValueSchedule** is now derived from **AbstractAtomicSchedule**
- Class **ScheduleComponent** is now derived from **AbstractADEFeatureWithLifeSpan**

Time series package:

- Added class **AbstractTimeSeriesFile**
- Added class **MonthlyTimeSeries**
- Added class **MonthlyTimeSeriesFile**
- Added class **TypicalValuesMonthlyTimeSeries**
- Added class **TypicalValuesMonthlyTimeSeriesFile**
- Added class **IrregularTimeSeries**
- Added class **IrregularTimeSeriesFile**
- Added class **TypicalValuesIrregularTimeSeries**
- Added class **TypicalValuesIrregularTimeSeriesFile**
- Renamed class **TypicalValuesTimeSeries** to **TypicalValuesRegularTimeSeries**
- Renamed class **TypicalValuesTimeSeriesFile** to **TypicalValuesRegularTimeSeriesFile**
- Class **RegularTimeSeries**:
 - Dropped property *temporalExtent*
 - Added property *startTimestamp*
 - Added property *endTimestamp*
- Class **RegularTimeSeriesFile**:
 - Class is now derived from class **AbstractTimeSeriesFile**
 - Dropped property *temporalExtent*
 - Dropped property *uom*
 - Dropped property *fileURI*
 - Dropped property *numberOfHeaderLines*
 - Dropped property *fieldSeparator*
 - Dropped property *recordSeparator*
 - Dropped property *decimalSymbol*
 - Dropped property *valueColumnNumber*
 - Added property *startTimestamp*
 - Added property *endTimestamp*
- Class **TypicalValuesRegularTimeSeriesFile**:
 - Class is now derived from class **AbstractTimeSeriesFile**
 - Dropped property *uom*
 - Dropped property *fileURI*
 - Dropped property *numberOfHeaderLines*
 - Dropped property *fieldSeparator*
 - Dropped property *recordSeparator*
 - Dropped property *decimalSymbol*
 - Dropped property *valueColumnNumber*

Version 3.0 beta 7

2025-10-14

No changes, except for version number going from 2.0 to 3.0.

Version 2.0 beta 7

2025-06-25**Schedules package:**

- Class **ScheduleLibrary**
 - Changed relation type of *libraryMember* to composition

Layered Construction package:

- Class **LayeredConstructionLibrary**
 - Changed relation type of *libraryMember* to composition
- Class **LayeredConstruction**
 - Added property *gValue*
- Class **MaterialLibrary**
 - Changed relation type of *libraryMember* to composition

2025-05-30**Schedules package:**

- Class **AtomicSchedule**
 - Changed multiplicity of relation *timeSeries* to [0..1]
 - Added property *constantValue*

2025-05-13**Building physics package:**

- Class **ThermalZone**
 - Changed relation type of *thermalBoundary* to composition

2025-05-07**Core package:**

- Class **ADE_CityObject**
 - Changed relation type of *device* to aggregation
 - Changed relation type of *weatherData* to composition
- Class **AbstractThermalZone**
 - Changed relation type of *usageZone* to association
- Class **AbstractUsageZone**
 - Changed relation type of *buildingUnit* to association
- Class **AbstractResource**
 - Made relation to **AbstractTimeSeries** explicit and set to composition
- Class **DeviceOperation**
 - Made relation to **AbstractSchedule** explicit and set to aggregation
- Class **WeatherData**
 - Made relation to **AbstractTimeSeries** explicit and set to composition

Occupancy package:

- Class **UsageZone**
 - Changed relation type of *occupiedBy* to aggregation
 - Made all relations to **AbstractSchedule** explicit and set to aggregation
- Class **BuildingUnit**
 - Changed relation type of *occupiedBy* to aggregation
- Class **Occupants**
 - Made relation to **AbstractSchedule** explicit and set to aggregation

Layered Construction package:

- Class **LayeredConstruction**
 - Changed relation type of *layer* to composition
- Class **Layer**
 - Changed relation type of *material* to aggregation

Schedules package:

- Class **ScheduleComponent**
 - Changed relation type of *scheduleComponentMember* to aggregation

2025-05-01**Core package:**

- Class **ADE _AbstractSchedule**
 - Moved from package Schedules to Core
- Class **ADE _AbstractTimeSeries**
 - Moved from package Time Series to Core

Occupancy package:

- Class **Occupants:**
 - Added property *averageDietType*
 - Added property *averageIncomeLevel*
 - Added property *averageInstructionLevel*

CodeLists package:

- Added class **DietTypeValue**
- Added class **InstructionLevelValue**
- Added class **IncomeLevelValue**

2025-04-23**CodeLists package:**

- Added values to class **AreaTypeValue**
- Renamed class **HeightReferenceValue** to **HeightTypeValue**
- Added class **SuitabilityReasonValue**
- Added class **UrbanSpaceTypeValue**
- Added class **UrbanSpaceEndUseValue**

DataTypes package:

- Added class **Suitability**

Core package:

- Class **ADE _CityObject**
 - Added property *suitability*
- Class **ADE _AbstractBuilding**
 - Renamed property *bdgFloorArea* to *bdgArea*
 - Renamed property *bdgHeightAboveGround* to *bdgHeight*
- Class **AbstractBuildingPartition**
 - Renamed property *floorArea* to *area*

Resources package:

- Added class **UrbanSpace**

2025-04-12**Core package:**

- Added class **AbstractLibrary**

Layered Construction package:

- Added class **LayeredConstructionLibrary**
- Added class **MaterialLibrary**

Schedules package:

- Added class **ScheduleLibrary**

CodeLists package:

- Class **FunctionInNetwork** renamed to **FunctionInNetworkValue**

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2025-02-06**DataTypes package:**

- Dropped class **HeatExchangeType**
- Dropped class **OpticalProperties**

Core package:

- Class **ADE_CityObject**:
 - Renamed property from *operationSchedule* to *deviceOperation*
- Class **AbstractDevice**:
 - Changed type of property *heatDissipation* to gml:Measure
 - Added property *heatDissipationConvectiveFraction*
 - Added property *heatDissipationLatentFraction*
 - Added property *heatDissipationRadiantFraction*

Occupancy package:

- Class **UsageZone**:
 - Renamed property *averageInternalGains* to *internalGains* and changed type to gml:Measure
 - Added property *internalGainsConvectiveFraction*
 - Added property *internalGainsLatentFraction*
 - Added property *internalGainsRadiantFraction*
- Class **Occupants**:
 - Changed type of property *heatDissipation* to gml:Measure
 - Added property *heatDissipationConvectiveFraction*
 - Added property *heatDissipationLatentFraction*
 - Added property *heatDissipationRadiantFraction*

Layered Construction package:

- Class **LayeredConstruction**:
 - Dropped property *opticalProperties*
 - Added property *glazingRatio*
 - Added property *emissivity*
 - Added property *reflectance*
 - Added property *transmittance*
- Class **Layer**:
 - Changed multiplicity of property *thickness* to [1]
- Class **AbstractMaterial**:
 - Added property *libraryCode*

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Resources package:

No changes since beta 4

Weather Station package:

No changes since beta 4

2025-01-30**Packages**

- Created TimeSeries package and moved all related classes into it
- Created Schedule package and moved all related classes into it
- Dropped Supporting classes package

CodeLists package:

- Class **CertificateTypeValue** renamed to **EnergyPerformanceCertificateTypeValue**

2025-01-28

CodeLists package:

- Added class **ScheduleTypeValue**

Enumerations package:

- Dropped class **DayTypeValue**

Supporting classes package:

- Dropped class **TimeSeriesSchedule**
- Dropped class **DailyPatternSchedule**
- Dropped class **PeriodOfYear**
- Dropped class **DailySchedule**
- Added class **AtomicSchedule**
- Added class **CompositeSchedule**
- Added class **ScheduleComponent**
- Class **AbstractSchedule**:
 - Added properties *type*, *startTime*, *startDay*, *startMonth*, *startYear*, *temporalExtent*

2025-01-22

Devices package:

- Class **AbstractSolarCollector**:
 - Added properties *azimuth* and *inclination*
 - Dropped property *surfaceGeometry*
 - Added properties *lod2MultiSurface* and *lod3MultiSurface*

2025-01-13

CodeLists package:

- Dropped class **ThermalBoundaryTypeValue**
- Added class **ShadingDeviceTypeValue**
- Added class **WeatherValueTypeValue**

DataTypes package:

- Moved class **PeriodOfYear** from package DataTypes to Supporting classes
- Moved class **DailySchedule** from package DataTypes to Supporting classes
- Class **AbstractQualifiedAttribute**:
 - Added properties *source* and *value*
- Classes **QualifiedHeight**, **QualifiedArea**, **QualifiedVolume**:
 - Dropped property *value*

Supporting classes package:

- Class **AbstractTimeSeries**:
 - Moved back from package Core to Supporting classes
 - Changed multiplicity of property *acquisitionMethod* to [1]
 - Changed multiplicity of property *interpolationType* to [1]
- Class **TypicalValuesTimeSeries**:
 - Added properties *startTime*, *startDay*, *startMonth*
- Class **TypicalValuesTimeSeriesFile**:
 - Added properties *startTime*, *startDay*, *startMonth*
- Added class **DualValueSchedule**:

- Class **PeriodOfYear**:
 - Changed stereotype from *dataType* to *featureType*
 - Added properties *startDay*, *startMonth*, *endDay*, *endMonth*
 - Dropped property *period*
- Class **DailySchedule**:
 - Changed stereotype from *dataType* to *featureType*
 - Added property *repetition*
 - Renamed property from *schedule* to *timeSeries*

Layered Construction package:

- Added class **ReverseLayeredConstruction**
- Dropped class **LayerComponent**
- Class **Layer**:
 - Removed property *layerComponent*
 - Added property *thickness*
 - Added property *material*
- Class **SolidMaterial**:
 - Renamed property from *specificHeat* to *specificHeatCapacity*
 - (Re)added and renamed property from *conductivity* to *thermalConductivity*
 - (Re)added property *embodiedEnergy*
 - (Re)added property *embodiedCarbon*
 - (Re)added property *porosity*

Core package:

- Class **AbstractDevice**:
 - Class is renamed from **AbstractEnergySystem** to **AbstractDevice**
 - Added property *heatDissipation*
 - Added property *operationSchedule*
 - Added property *efficiencyIndicator*
 - Added property *installedPower*
 - Added property *nominalEfficiency*
- Class **AbstractTimeSeries**:
 - Moved back from package Core to Supporting classes
- Class (ADE) **_CityObject**:
 - Renamed property from *energySystem* to *device*
- Class **OperationSchedule**:
 - Moved from package Devices to Core
- Class **WeatherData**:
 - Renamed property from *timeseries* to *timeDependentValues*
 - Renamed property from *singleValue* to *yearlyValue*
 - Renamed property from *singleValueType* to *valueType*
 - Changed type of property *valueType* from *CharacterString* to *CodeList*
- Added class **AbstractBuildingPartition**
- Classes **AbstractBuildingUnit**, **AbstractUsageZone**, **AbstractThermalZone**
 - Changed parent class

Building physics package:

- Dropped class **ThermalBoundary**
- Dropped class **ThermalOpening**
- Class **ThermalZone**
 - Removed property *floorArea* (now inherited via **AbstractBuildingPartition**)
 - Removed property *volume* (now inherited via **AbstractBuildingPartition**)
 - Removed property *volumeGeometry* (now inherited via **AbstractBuildingPartition**)
 - Removed property *boundedBy*
 - Added property *heatCapacity*
 - Added property *coincidesWithLod2Hull*
 - Added property *coincidesWithLod3Hull*
 - Added property *thermalBoundary*
- Added class **PartyWallSurface**

- Added class **BasementCeilingSurface**
- Added class **IntermediateFloorSurface**
- Added class **AtticFloorSurface**
- Class (ADE) **BoundarySurface**:
 - Renamed property from *bdgBdrySrfWindowToWallRatio* to *bdgBdrySrfOpeningToSurfaceRatio*
 - Renamed property from *bdgBdrySrfArea* to *bdgBdryTotalSurfaceArea*
 - Added property *bdgBdryOpaqueSurfaceArea*
 - Added property *bdgBdrySrfThickness*
 - Added property *bdgBdrySrfHeatCapacity*

Occupancy package:

- Dropped class **AbstractFacilities**
- Dropped class **DHWFacilities**
- Dropped class **LightingFacilities**
- Dropped class **ElectricalAppliances**
- Dropped class **GenericFacilities**
- Class **BuildingUnit**:
 - Removed property *floorArea* (now inherited via **AbstractBuildingPartition**)
 - Removed property *volume* (now inherited via **AbstractBuildingPartition**)
 - Removed property *equippedWith*
- Class **UsageZone**:
 - Removed property *floorArea* (now inherited via **AbstractBuildingPartition**)
 - Removed property *volume* (now inherited via **AbstractBuildingPartition**)
 - Removed property *volumeGeometry* (now inherited via **AbstractBuildingPartition**)
 - Removed property *equippedWith*

Devices package:

The former EnergySystems package is renamed to Devices package

- Class **OperationSchedule**:
 - Moved from package Devices to Core
- Dropped class **AbstractEnergyConversionSystem**
 - Now all subclasses are derived directly from **AbstractDevice**
- Dropped class **AbstractStorageSystem**
 - Now all subclasses are derived directly from **AbstractDevice**
- Class **GenericDevice**:
 - Renamed from class **GenericEnergyConversionSystem**
- Class **ElectricalStorageDevice**:
 - Renamed from class **PowerStorageSystem**
- Class **ThermalStorageDevice**:
 - Renamed from class **ThermalStorageSystem**
- Class **AbstractSolarCollector**:
 - Renamed from class **AbstractSolarEnergySystem**
- Class **PhotovoltaicThermalCollector**:
 - Renamed from class **PhotovoltaicThermalSystem**
- Class **PhotovoltaicCollector**:
 - Renamed from class **PhotovoltaicSystem**
- Class **SolarThermalCollector**:
 - Renamed from class **SolarThermalSystem**
- Class **GenericSolarCollector**:
 - Renamed from class **GenericSolarEnergySystem**
- Added class **LightingDevice**
- Added class **GenericElectricalDevice**
- Added class **MovableShadingDevice**

Urban Function Areas package:

- Class **UrbanFunctionArea**:
 - Removed property *rank*
-

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2024-09-25

- Codelist **ThermalBoundaryTypeValue**
 - Renamed *outerWall* to *exteriorWall*
 - Renamed *intermediaryFloor* to *intermediateFloor*

2024-09-20

- Minor edits to the layout of the UML diagrams
- Polished Codelist **ThermalBoundaryTypeValue**

2024-08-12

- Class **ADE CityObject**:
 - Renamed association *connectedTo* to *utilityNetworkConnection*
 - Remodelled association class **CityObjectRelation**
- Class **AbstractTimeSeries**:
 - Moved to Core module
- Class **WeatherData**:
 - renamed property *valuesList* to *timeSeries*
 - made the association to **AbstractTimeSeries** explicit
- Added some values to codelist **RelationTypeValue**
- Class **TimeSeriesSchedule**:
 - Changed association to **AbstractTimeSeries** to Composition
- Class **DailySchedule**:
 - Changed association to **AbstractTimeSeries** to Composition
- Class **SensorConnection**:
 - Changed type of *uom* to UnitOfMeasure
 - Renamed *linkToSensorDescription* to *linkToSensorDescription*
- Class **AbstractSolarEnergySystem**:
 - Removed association *installedOn* (now taken over by **CityObjectRelation**)
- Class **Energy**:
 - Changed multiplicity of *EnergyCarrier* to [0..1]
- Class **Waste**:
 - Changed multiplicity of *isDanerous* and *isRecyclable* to [0..1]

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